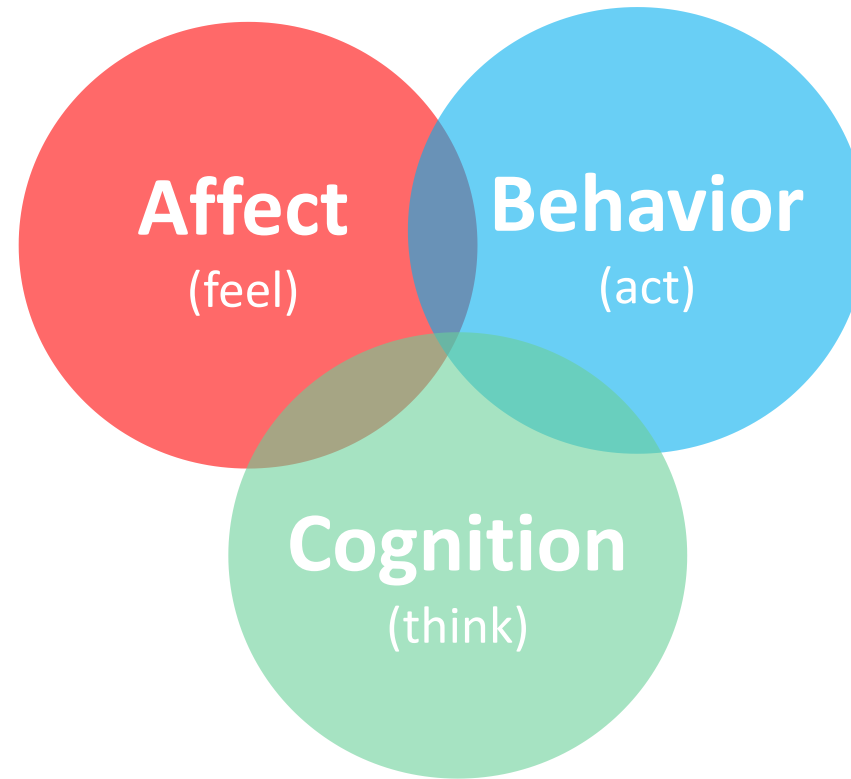


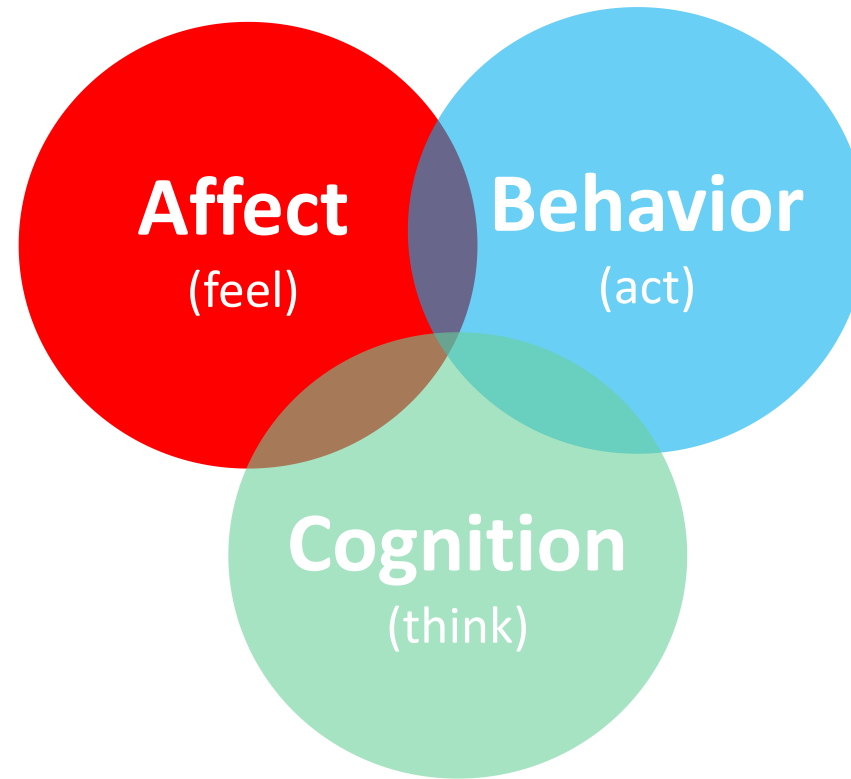


Artificial Intelligence for Digital Characters

Affective Computing

Dr. Rafael Wampfler, ETH Zürich





What is Affect?

Emotions

Angry

Sad

Fearful

Moods

Relaxed

Bored

Interpersonal stances

Distant

Warm

Supportive

Preferences & Attitudes

Liking

Hating

What is Affect?

Emotions

Angry

Sad

Fearful

Moods

Relaxed

Bored

Interpersonal stances

Distant

Warm

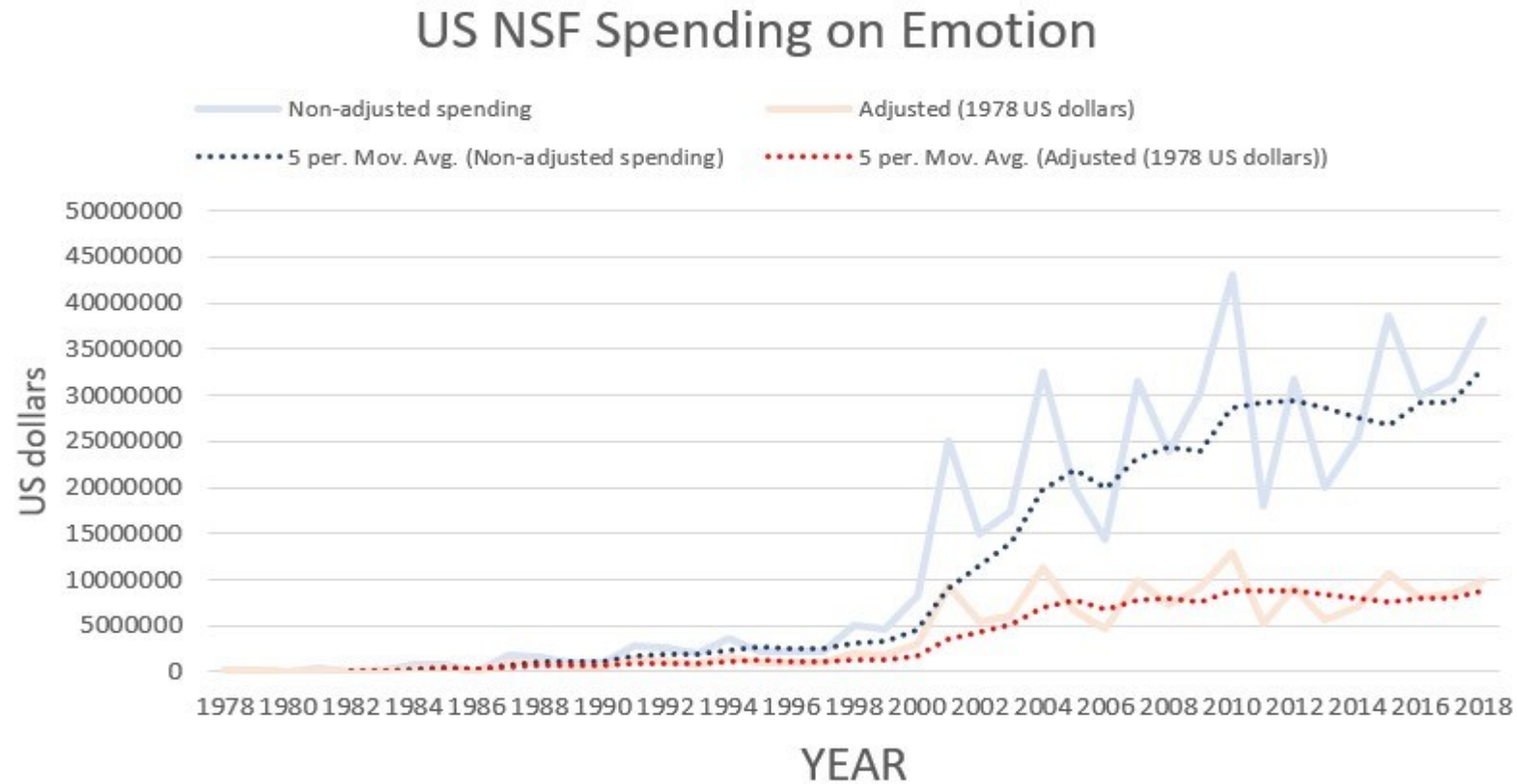
Supportive

Preferences & Attitudes

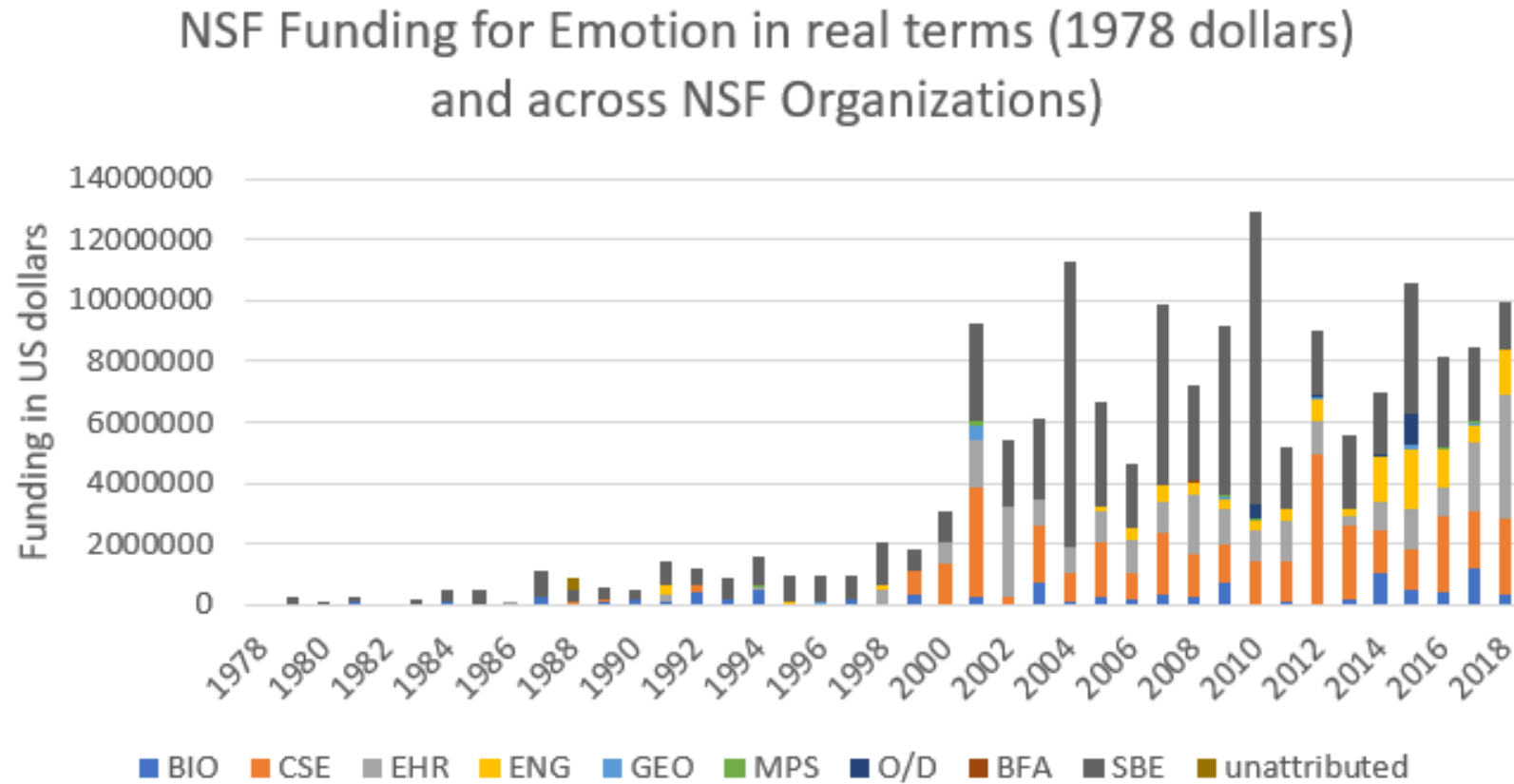
Liking

Hating

Are We in an Era of Affectivism?



Are We in an Era of Affectivism?



Affective states are important in education...

... Influence learning gain

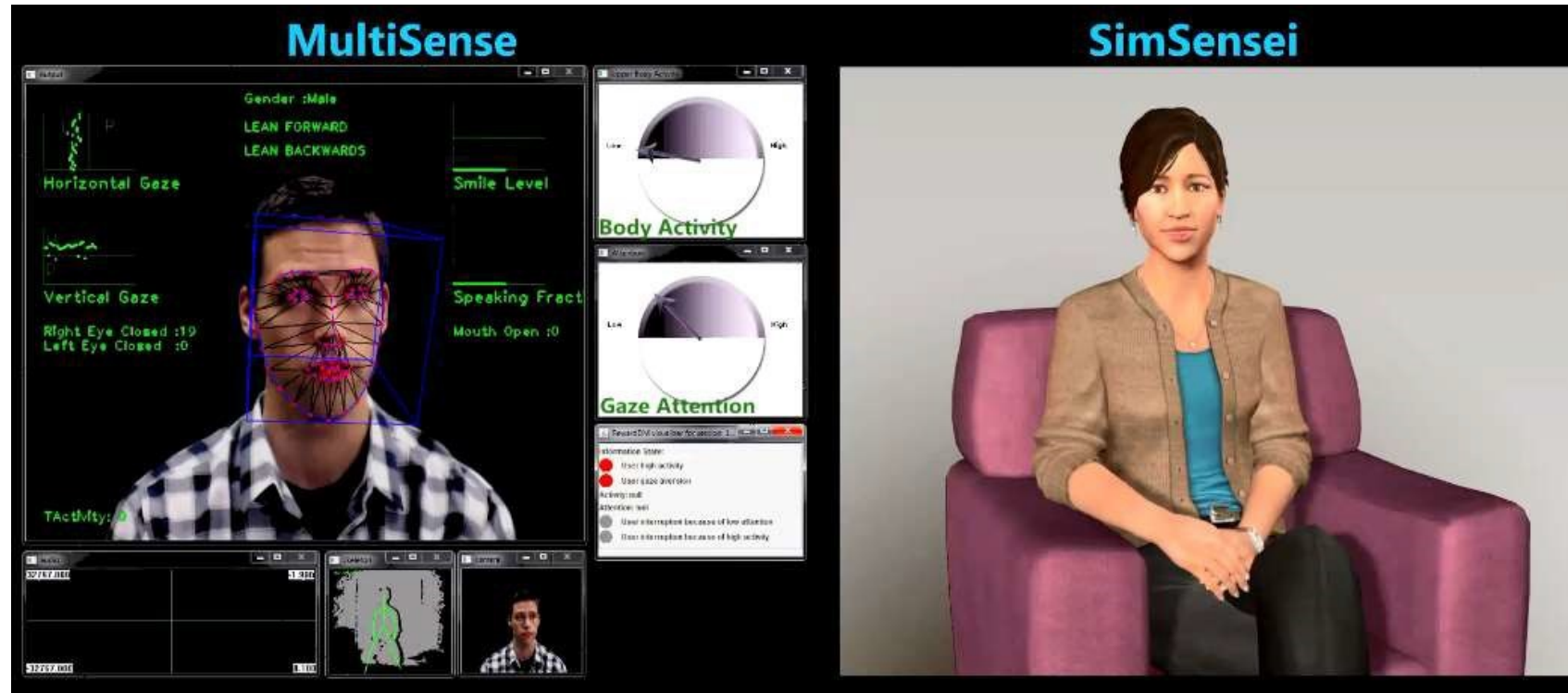
Affective states are important in education...

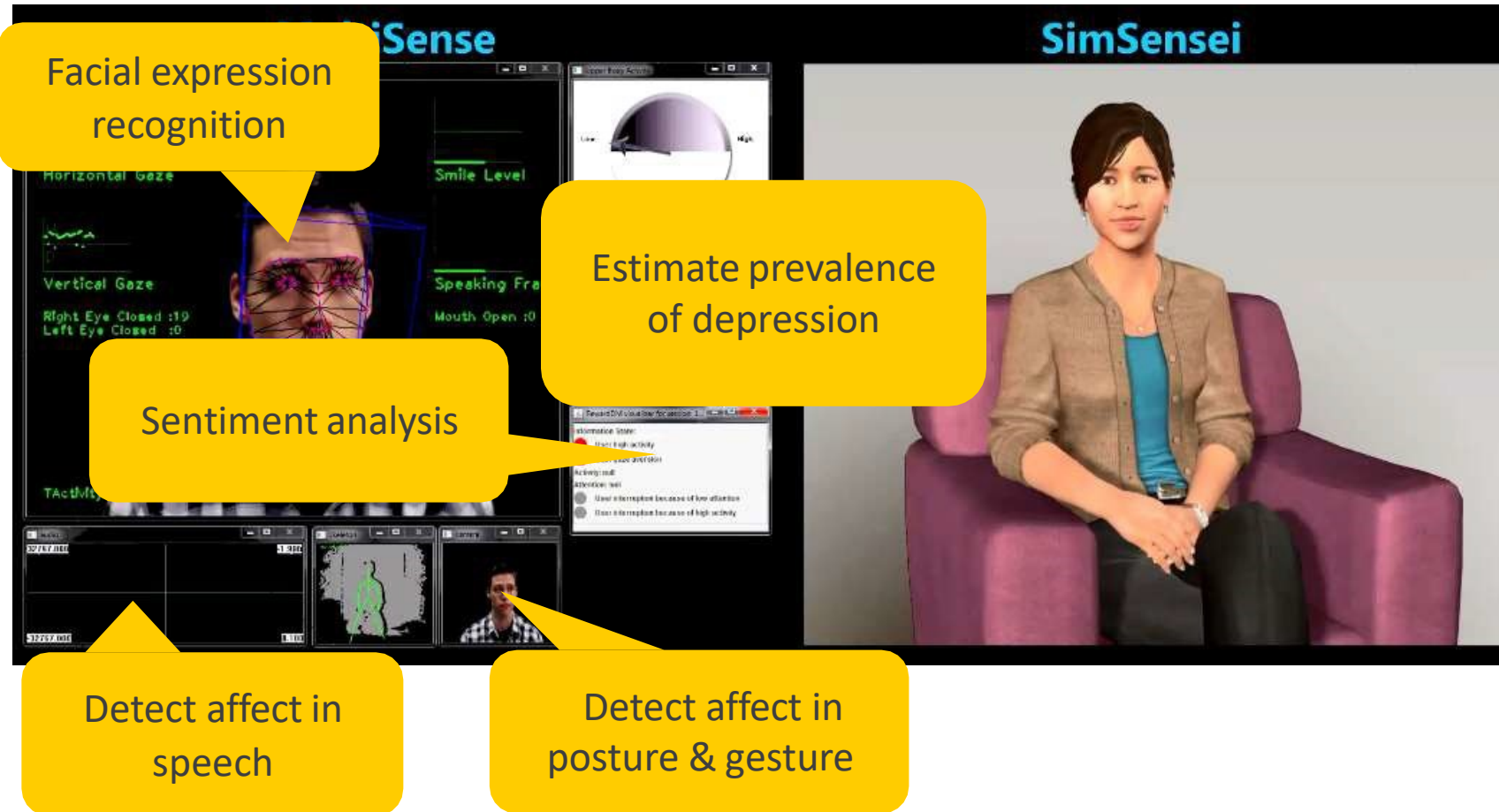
... Influence learning gain

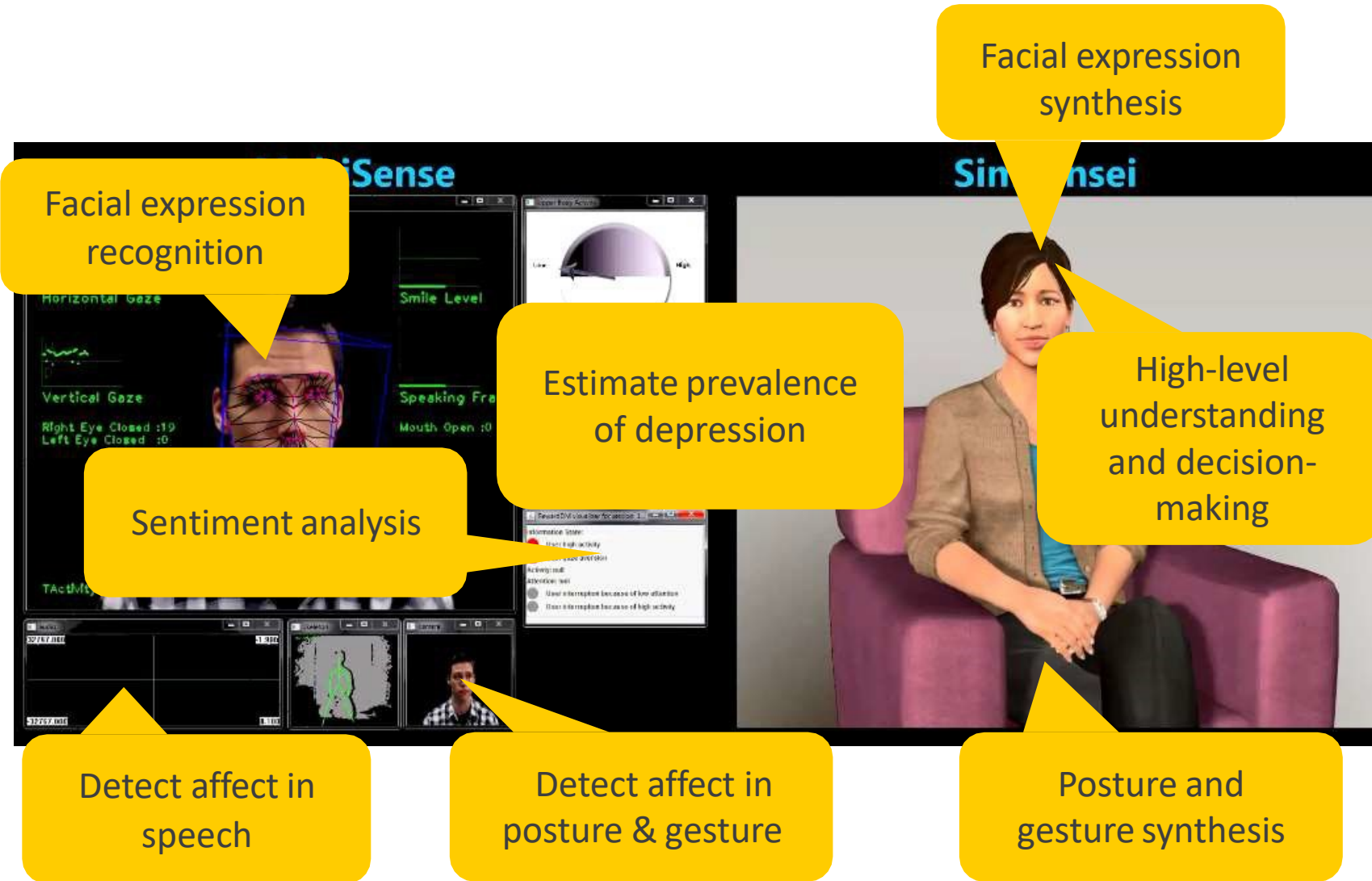


Predict affective state

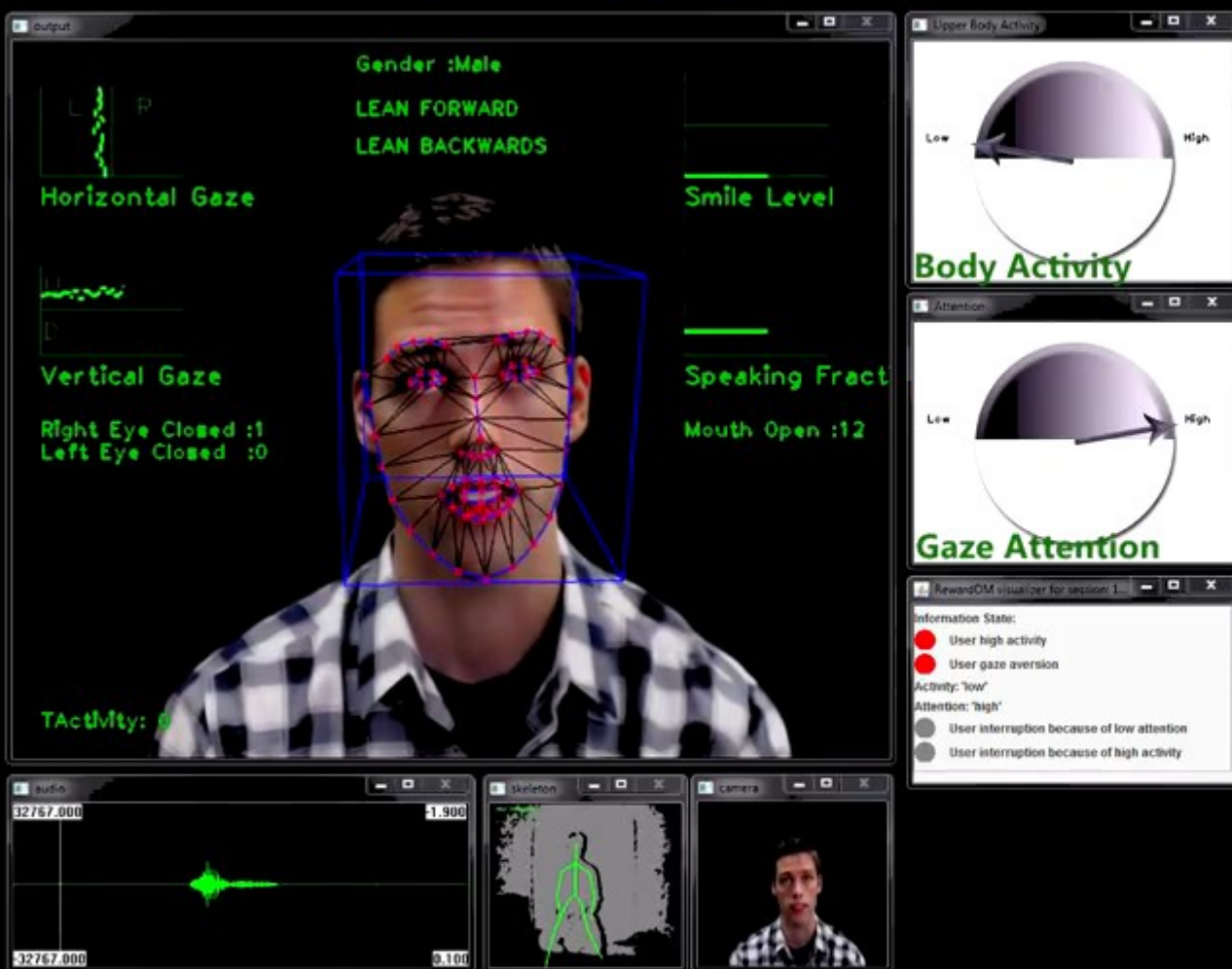
... Remedial interventions







MultiSense



SimSensei



SimSensei nods at appropriate points in the dialogue.



Incoming

New ▾ Reply ▾ Reply to All ▾ Forward ▾			Report as Spam More ▾	🔍	🗨
Who	Emotions	Subject			
Michal Shmueli-Sc	😊	collaboration plans			
Michal Shmueli-Sc	😊	what do you think?			
Michal Shmueli-Sc	😊	Song Analysis			
Michal Shmueli-Sc	😊	Optional Candidate			
Guy Feigenblat		Watson Platform Next			
Guy Feigenblat		Watson Platform Next			
Orith Toledo-Roner		Re: my address			
kobys		Usb computer			
Scott Duby		Rescheduled: Discuss Affective Computing (Nov 6 10:00 AM CST in Audio Conference Access Numbers)			
root		AFS/GPFS password of shmueli is not changed for more than 281 days			
Benjamin Sznajder		Data analysis results			
Scott Duby		Re: Discuss Incorporating Effective Computing into NRF Scenario			
David Konopnicki		Re: nViso Emotion Analytics Demo NRF + H2020 ICT Calls			
David Konopnicki		Re: Fw: Discuss Incorporating Effective Computing into NRF Scenario			
Michal Shmueli-Sc		File			

Optional Candidate
Michal Shmueli-Scheuer to: Michal Shmueli-Scheuer

Hi Michal,
I met John in last SIGIR. Seems a nice guy with relevant background. I gave his CV to Alice once.
Lets bring him for interview.

Outgoing

Send Send and File... Save as Draft Delivery Options... Signature ▾ Display ▾ More ▾

To: Guy Feigenblat/Haifa/IBM@IBMIL

Cc:

Bcc:

Subject:

Hi Guy,

i had a really bad experience at the office today.my boss shout at me. hate that going to have a drink now

See you later,

Thanks,
Michal

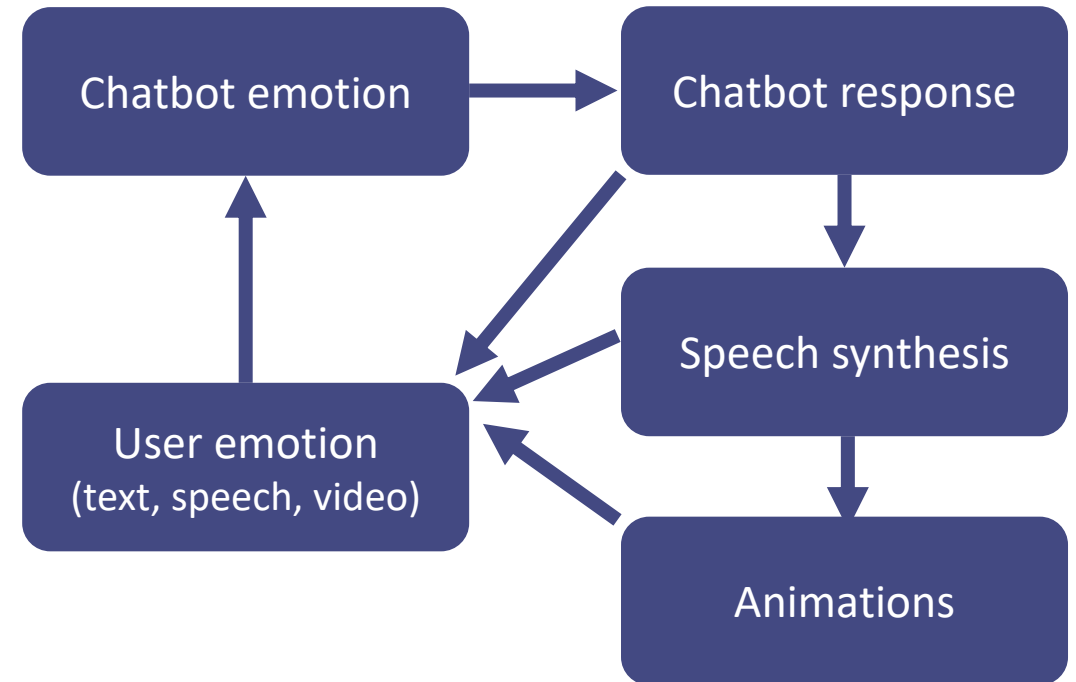
Michal Shmueli-Scheuer, PhD
Information Retrieval
IBM Haifa Research Labs
Phone: +972-3-768-9419 Fax: +972-3-768-9545
Email: shmueli@il.ibm.com

Summerized Emotions

anger-10 sadness-6

List of emotional words:
bad - sadness
shout - sadness
hate - anger

OK



Theories

Video

Biosensors

James-Lange Theory
Cannon-Bard Theory
Schachter-Singer Theory
Dimensional
Basic Emotions

Components of Emotions

Subjective experience

Psychological changes

Autonomic
Nervous System

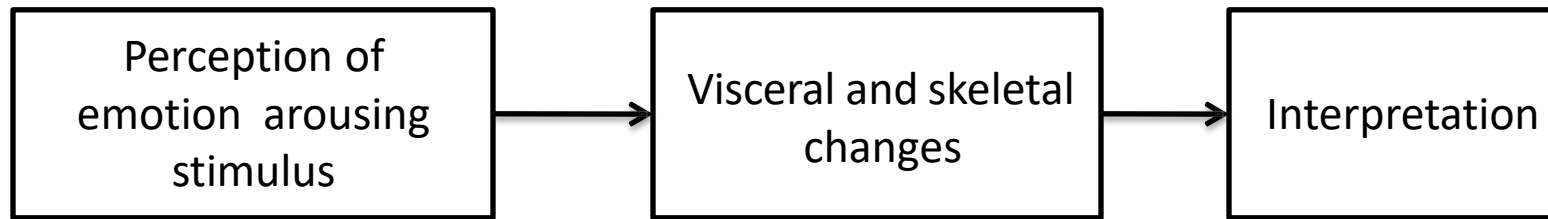
Endocrine
System

Behavior evoked

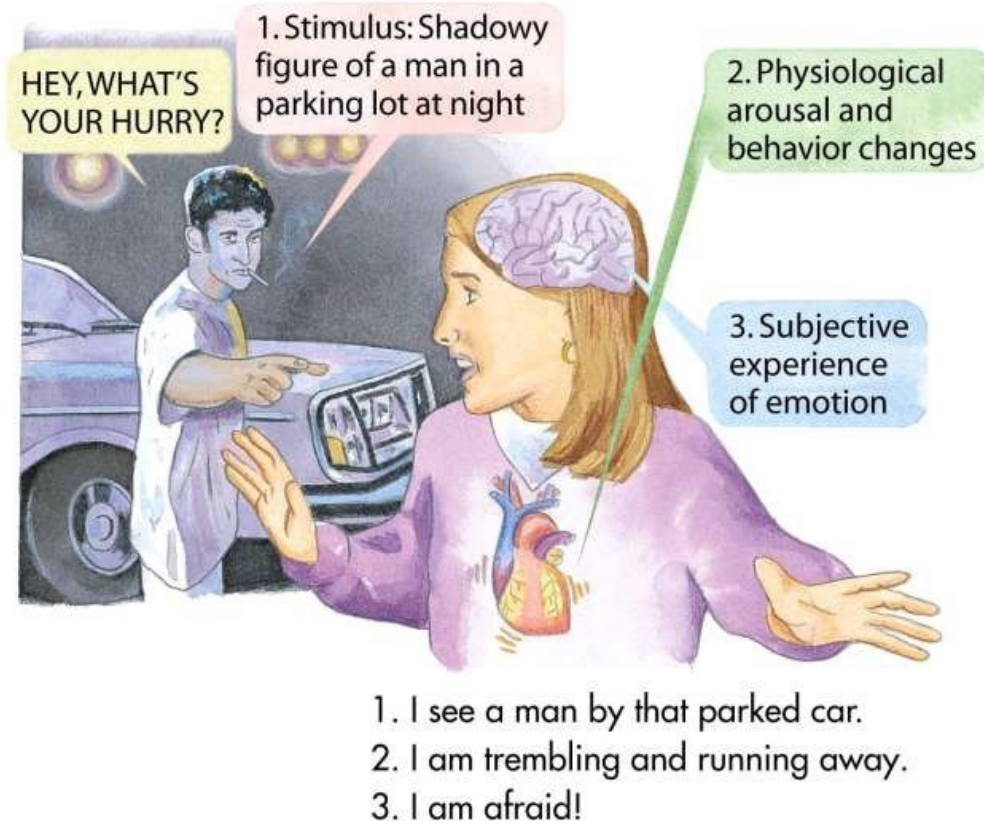
What is the time sequence of these components in an emotion episode?



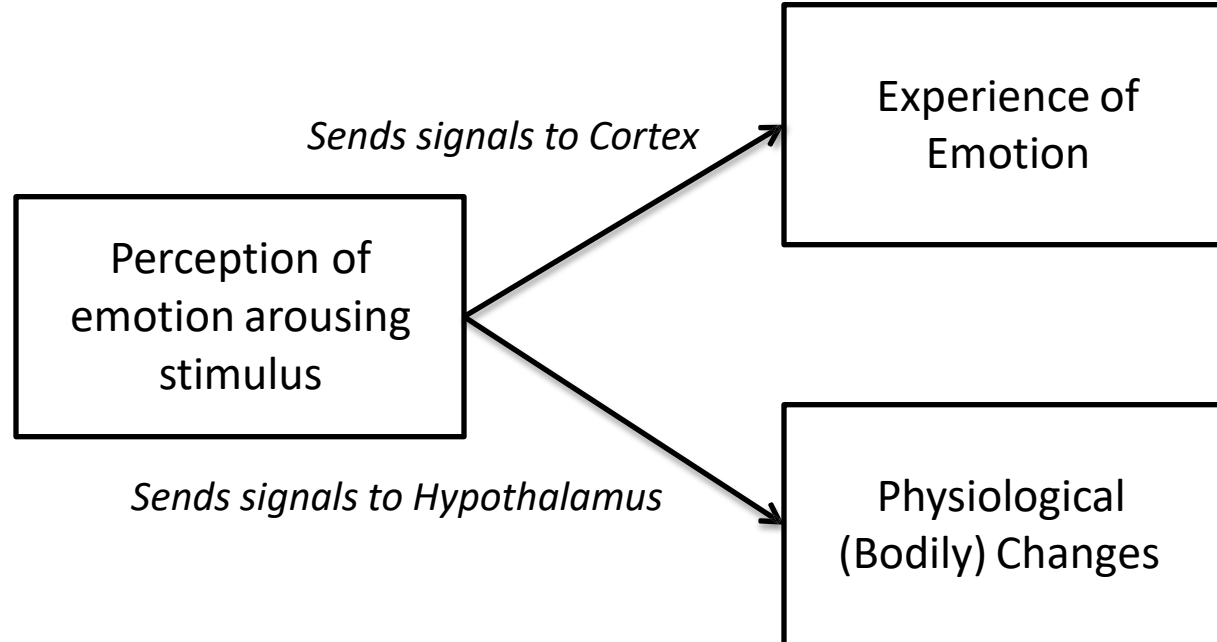
James-Lange Theory



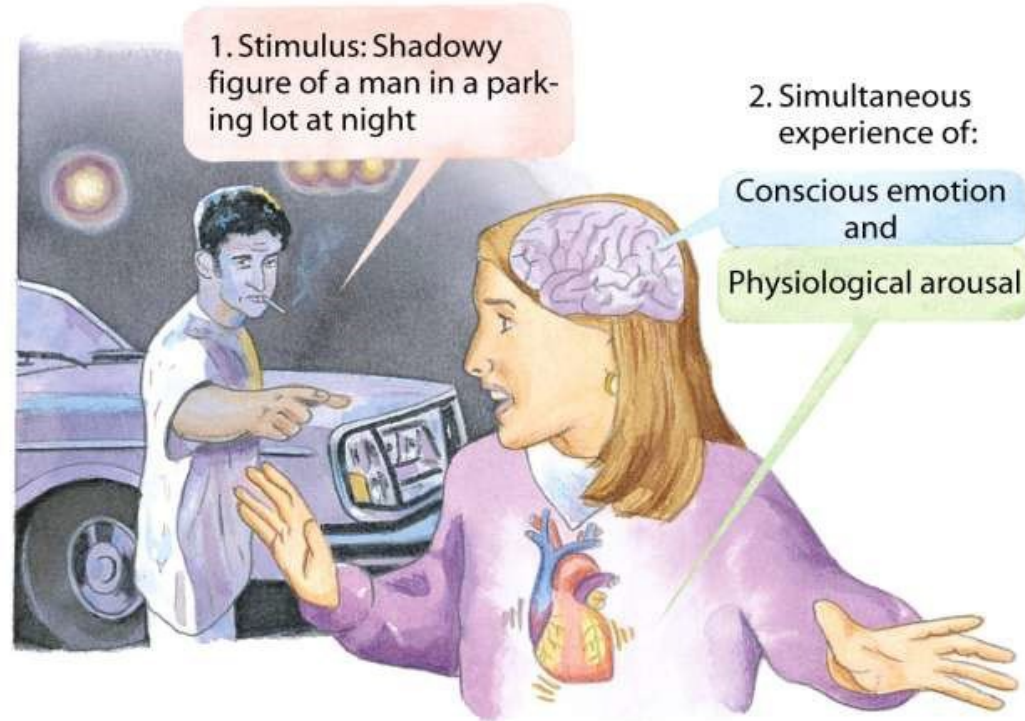
James-Lange Theory



Cannon-Bard Theory

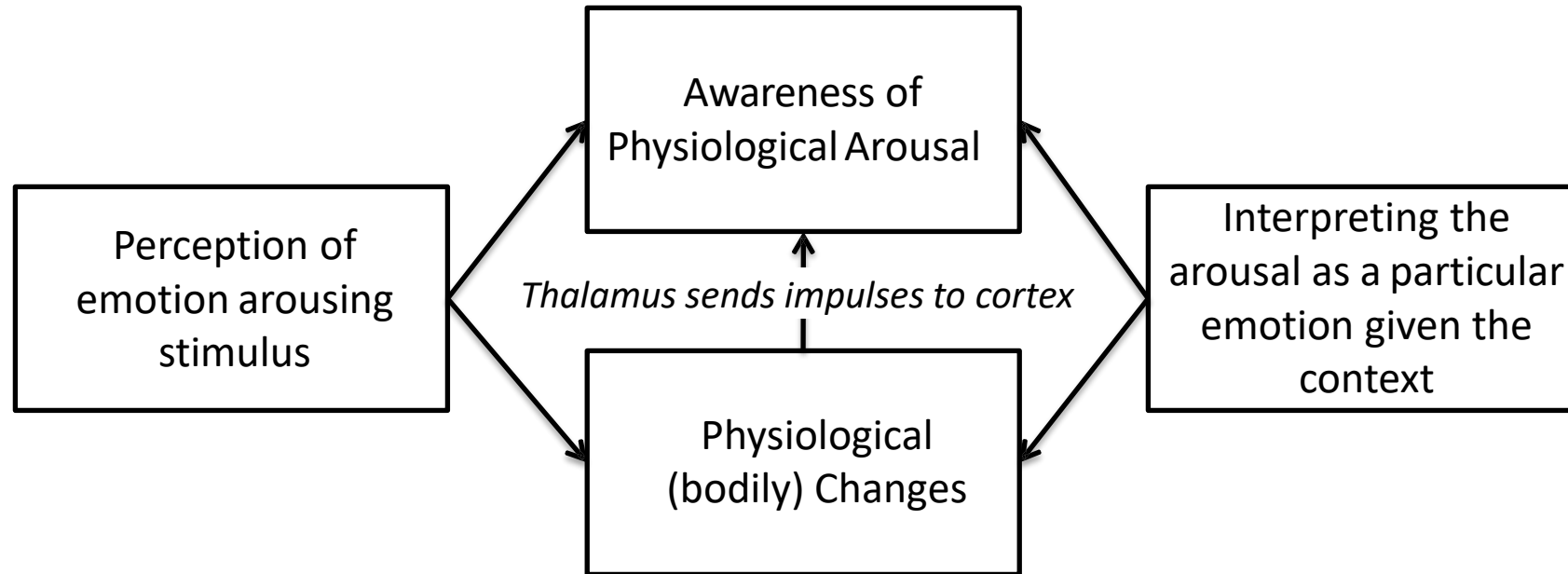


Cannon-Bard Theory

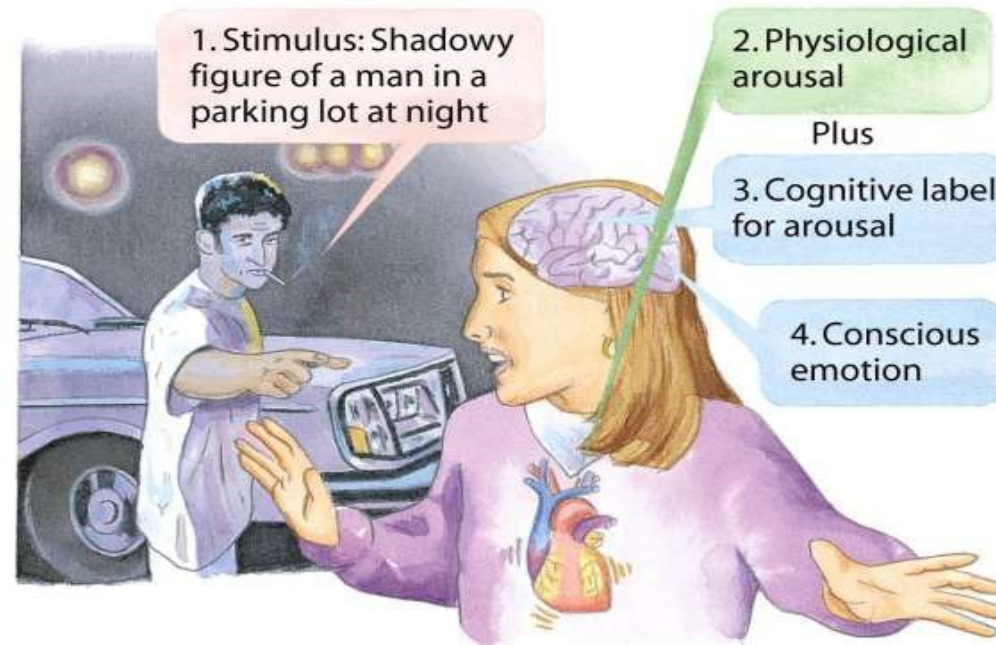


1. I see a man by that parked car.
2. I am afraid *and* trembling!

Schachter-Singer Theory



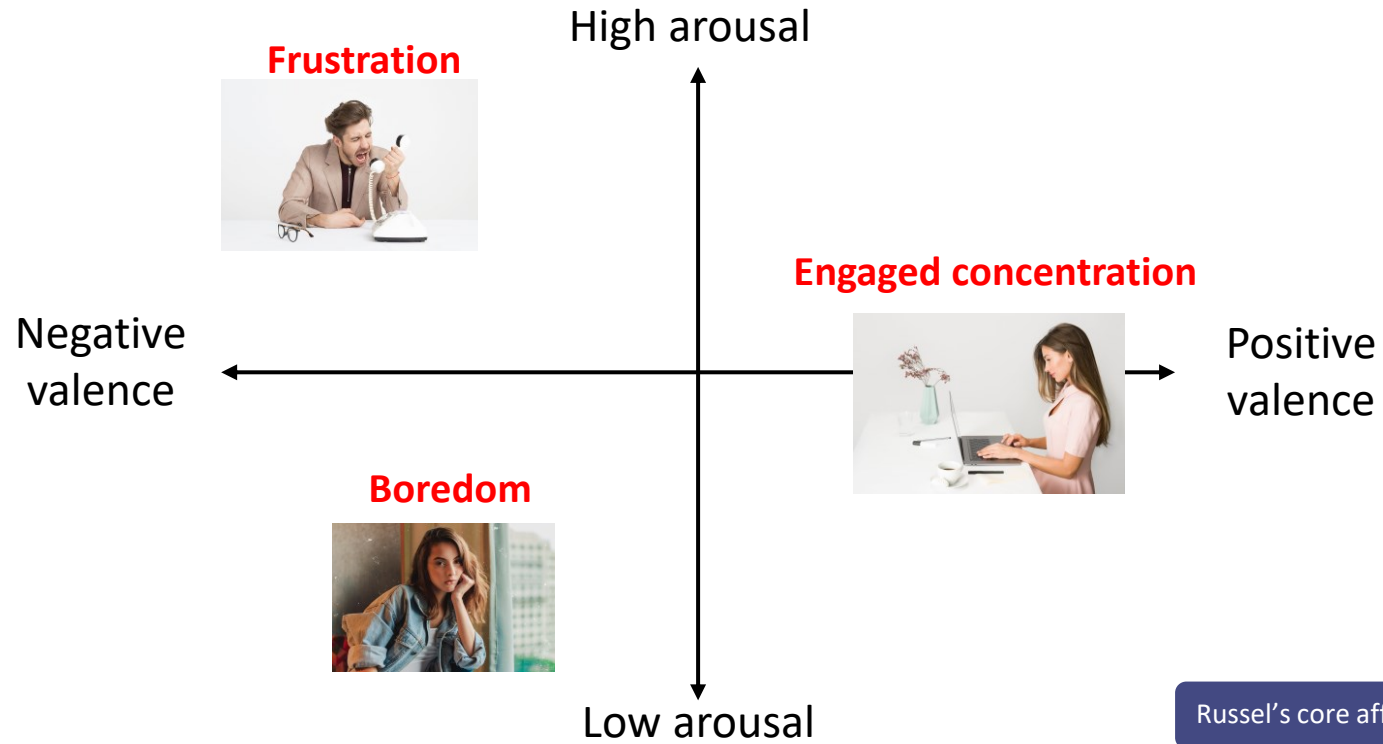
Schachter-Singer Theory



1. I see a man by that parked car.
2. I am trembling.
3. My trembling is caused by fear.
4. I am afraid!

- 1) Core affect: Most elementary affective feelings
Part of a full-blown emotional episode
- 2) Valence = Positive or negative feeling
Arousal = Intensity of feeling
Dominance = Degree of control exerted by the emotion

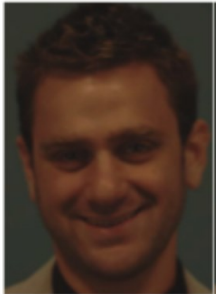
Dimensional Theories



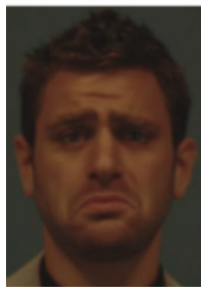
- 1) Caused by the perception of phylogenetic categories of events
Innate events such as perceiving a dangerous animal (bear)
- 2) Eliciting internal and external signals
Physiological changes and facial expressions
- 3) Trigger innate hard-wired action plans
Fight or flight

Ekman's basic emotions

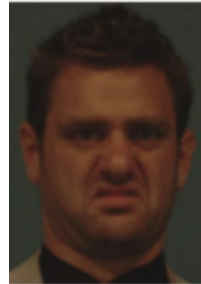
Joy



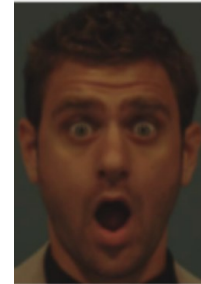
Sadness



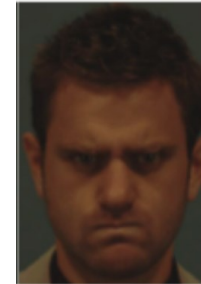
Disgust



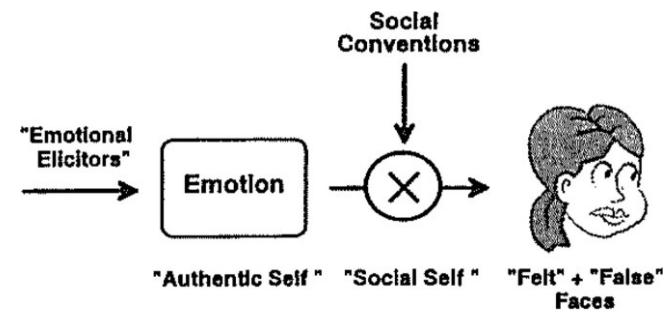
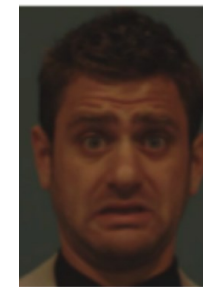
Surprise



Anger

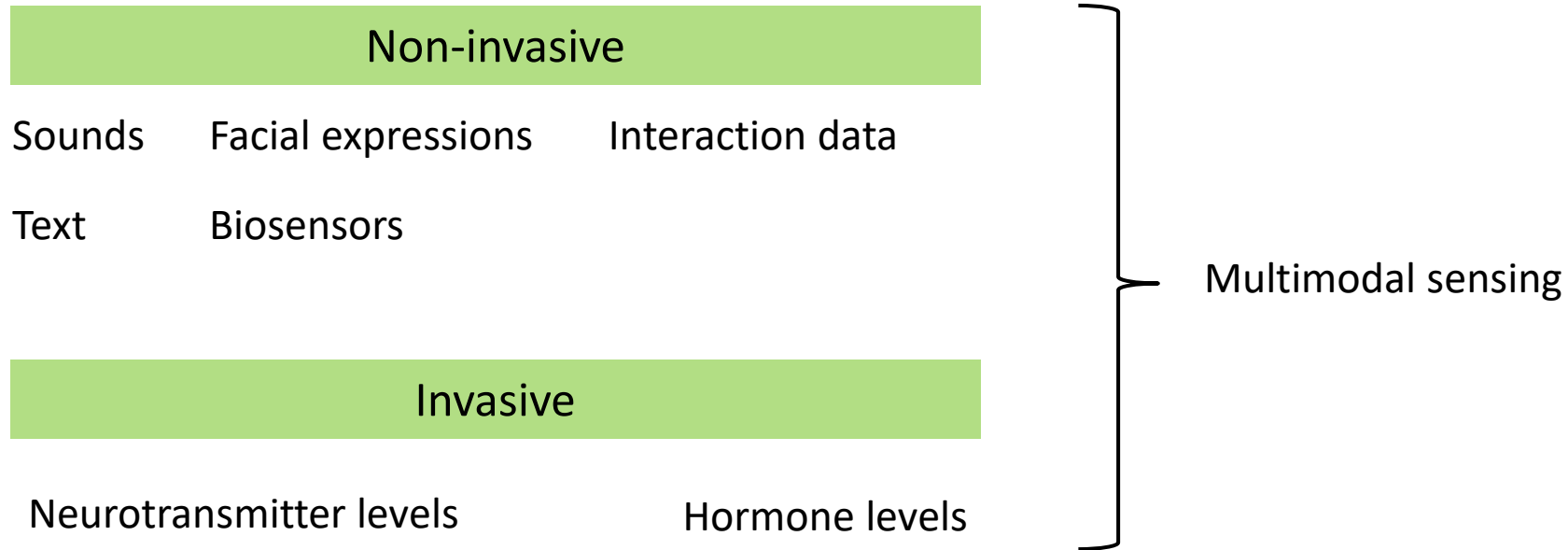


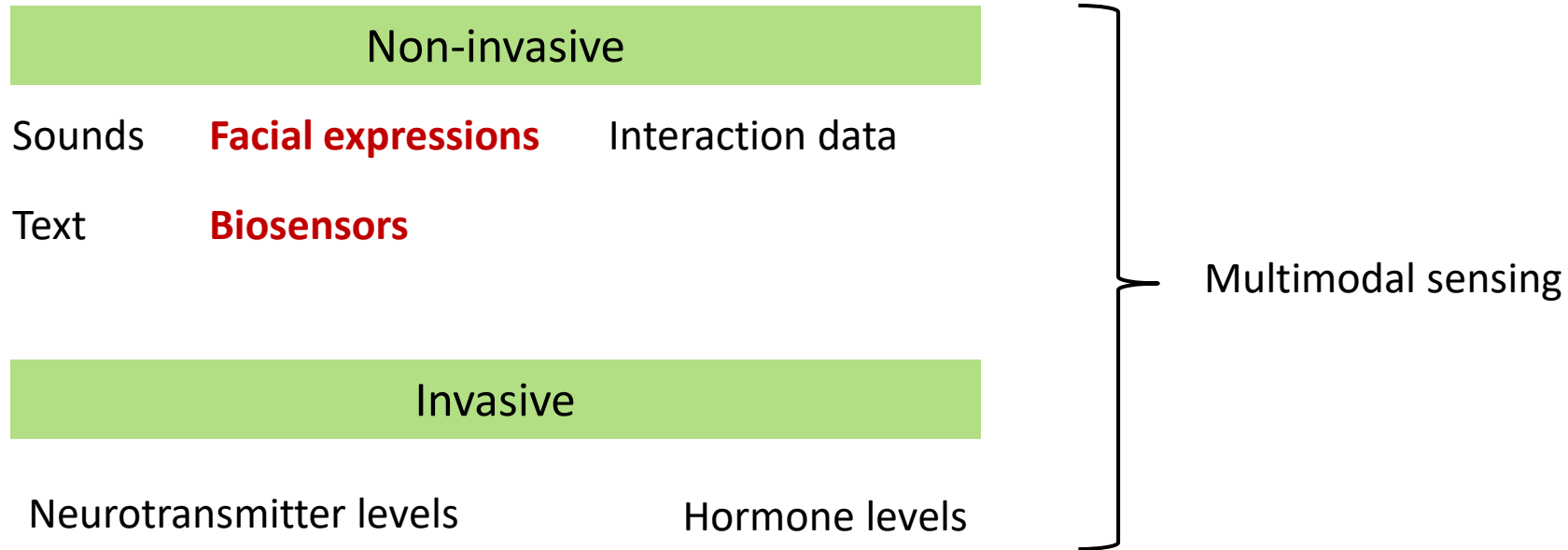
Fear

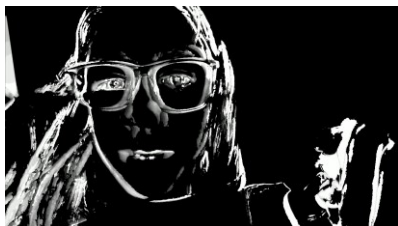
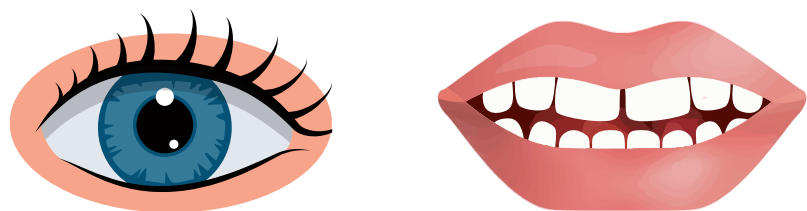


Plutchik Emotion Wheel





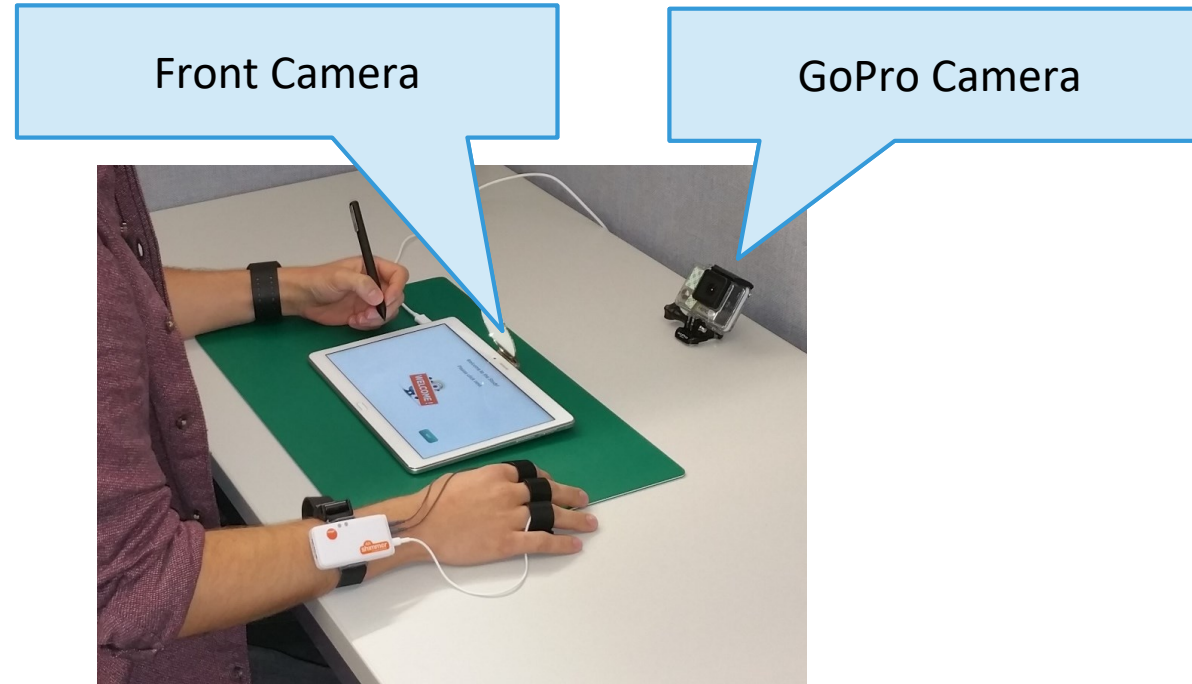


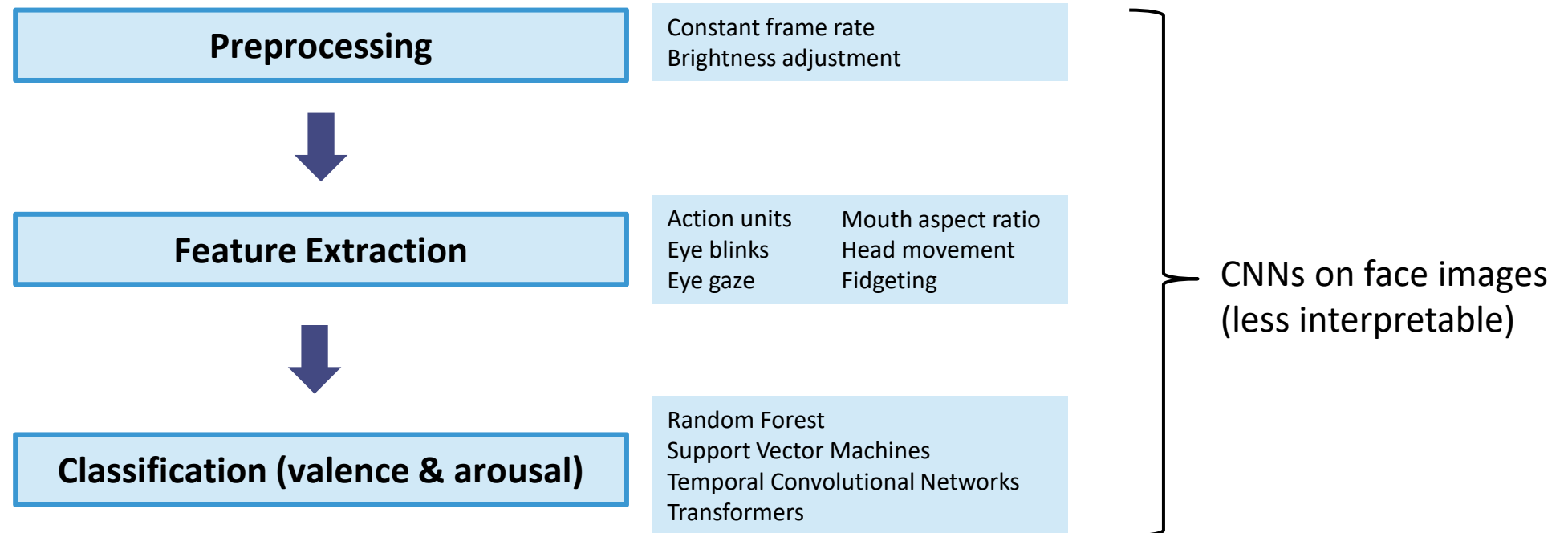


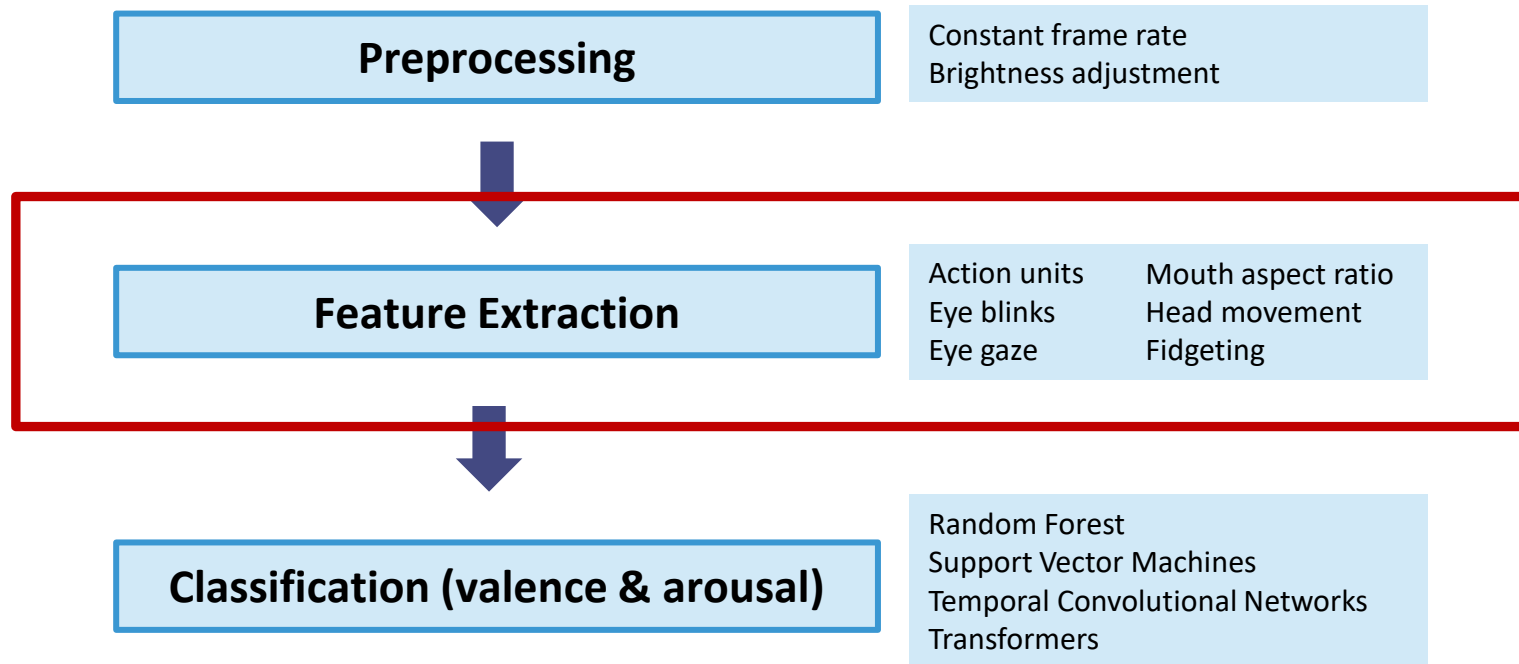
Theories **Video** Biosensors

Action units
Eye blinks
Eye gaze
Mouth aspect ratio
Fidgeting

Camera Recordings









46 Facial Action Units

AU1: Inner Brow Raiser

AU5: Upper Lid Raiser

AU26: Jaw Drop

<https://imotions.com/blog/facial-action-coding-system/>

Identify independent motions of the face

Regions: Eye, cheek, nose, mouth, chin

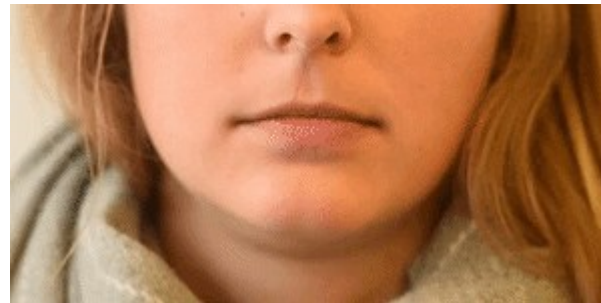
AU1 (Inner Brow Raiser)



AU5 (Upper Lid Raiser)



AU26 (Jaw Drop)



Action Units

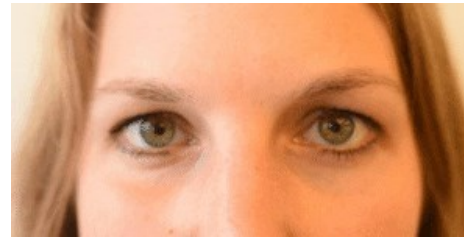
Basic Emotion	Action Units	Description
Happiness	6 + 12	Cheek Raiser, Lip Corner Puller
Sadness	1 + 4 + 15	Inner Brow Raiser, Brow Lowerer, Lip Corner Depressor
Surprise	1 + 2 + 5 + 26	Inner Brow Raiser, Outer Brow Raiser, Upper Lid Raiser, Jaw Drop
Fear	1 + 2 + 4 + 5 + 7 + 20 + 26	Inner Brow Raiser, Outer Brow Raiser, Brow Lowerer, Upper Lid Raiser, Lid Tightener, Lip Stretcher, Jaw Drop
Anger	4 + 5 + 7 + 23	Brow Lowerer, Upper Lid Raiser, Lid Tightener, Lip Tightener
Disgust	9 + 15 + 16	Nose Wrinkler, Lip Corner Depressor, Lower Lip Depressor

Surprise: Action Units 1 + 2 + 5 + 26

Inner Brow Raiser



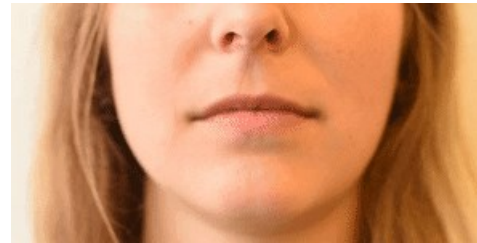
Outer Brow Raiser



Upper Lid Raiser



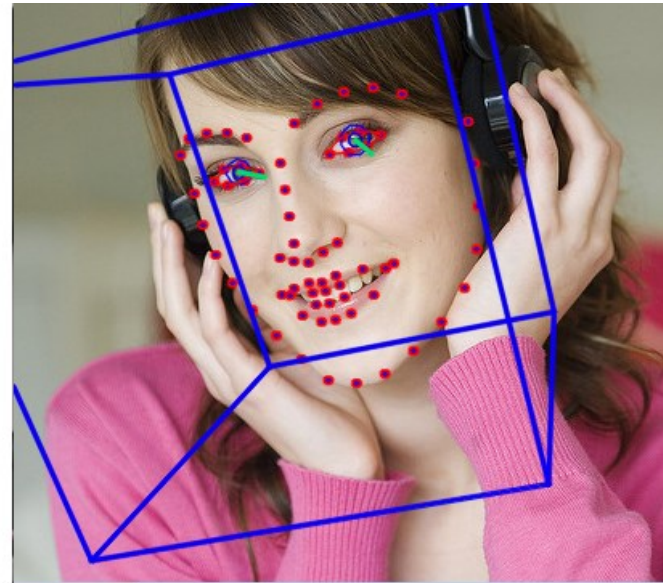
Jaw Drop



68 facial landmarks
Intensity of action units [0,5]
Presence of action units (present vs. absent)

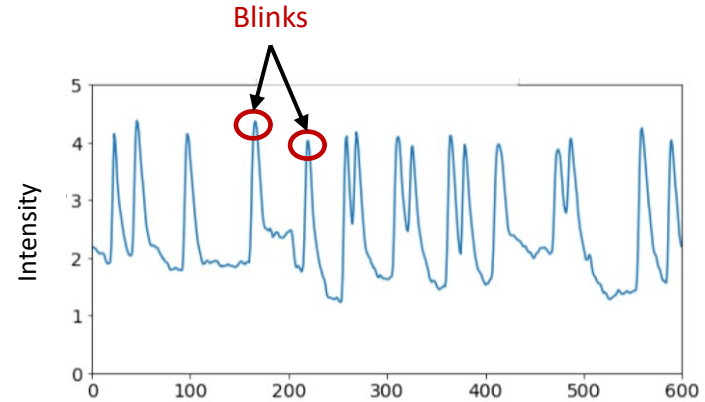


Eye gaze
Head Pose

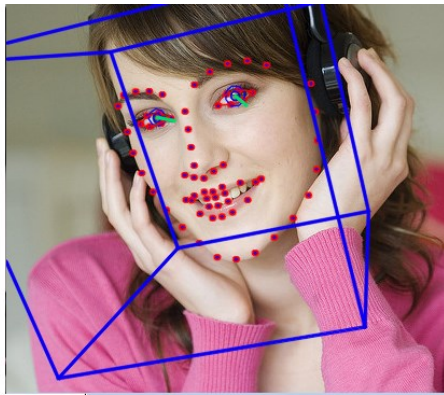


<https://github.com/TadasBaltrusaitis/OpenFace>

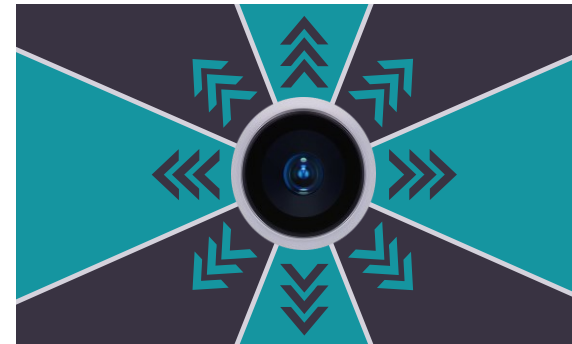
Based on AU 45 (eye blink) evaluated per frame



Individuals might look away when looking at emotionally disturbing pictures

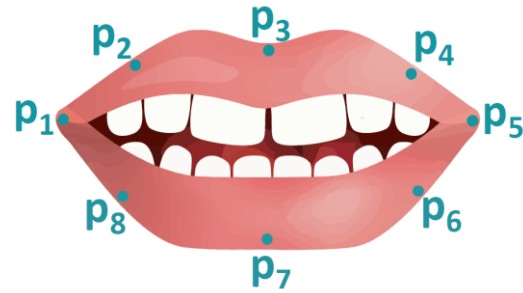


9 gaze regions



Mouth Aspect Ratio

Based on facial landmarks



$$MAR = \frac{\|p_2 - p_8\| + \|p_3 - p_7\| + \|p_4 - p_6\|}{3 * \|p_5 - p_1\|}$$

Captures all movement in the video (body, face,...)

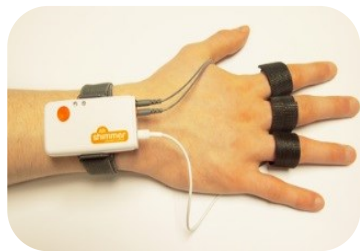


1. $f_{temp} = f_{gray} - b_{gray}$
2. *Binarizing f_{temp} (thresholding)*
3. *Energy E = Percentage of surviving pixels*
4. $b_{gray} = (1 - a) * b_{gray} + a * f_{gray}$

Basic statistics (mean, std, min, max,...)

Sum over all frames

Position of frame with min(E) and max(E)



Biosensors

Theories
Video

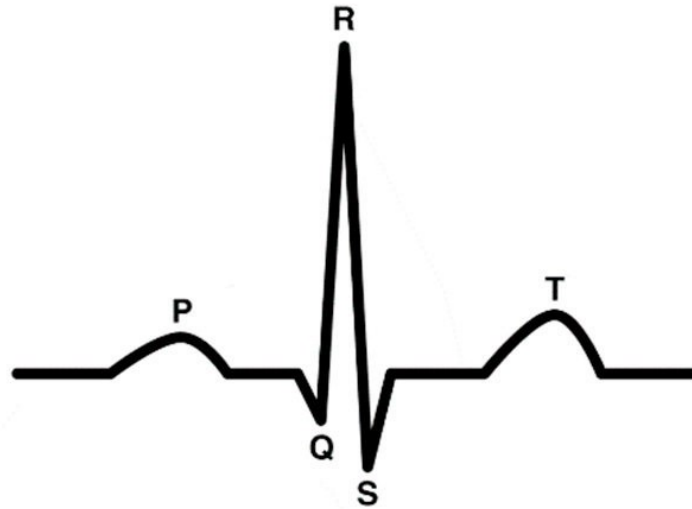
Heart rate variability
Skin conductance
Artifact detection

Cardiac Activity

ECG: Trace of electrical activity captured from the surface of the skin

PPG: Measuring amount of light reflected by vessels

Webcam: Measuring PPG signal from pupil and skin

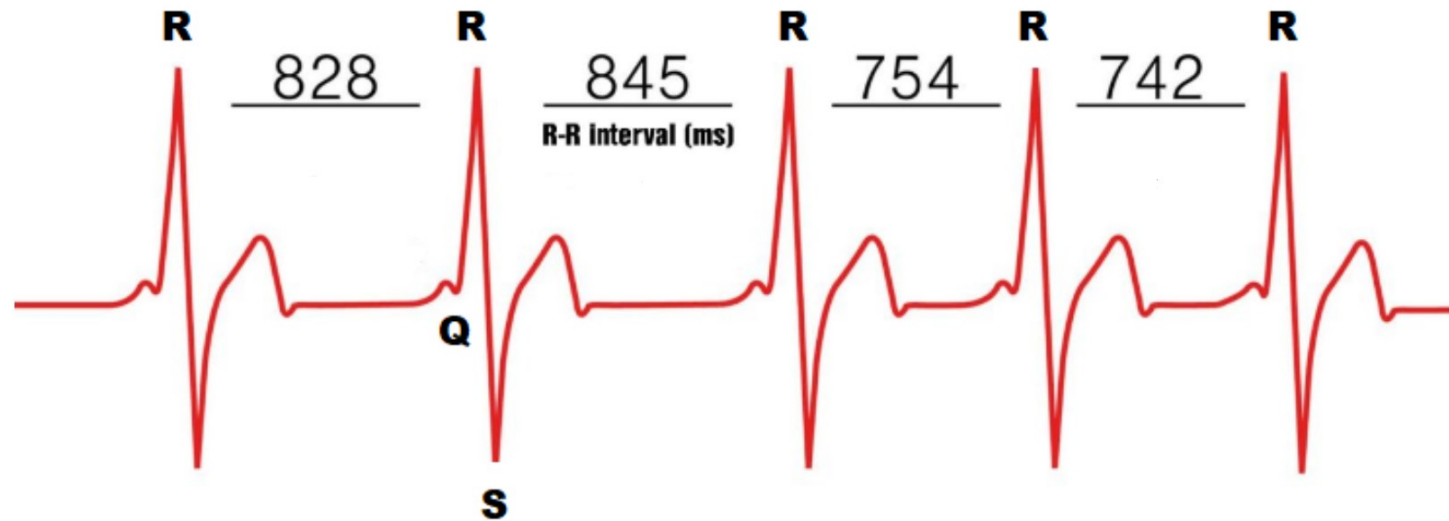


Cardiac Activity

ECG: Trace of electrical activity captured from the surface of the skin

PPG: Measuring amount of light reflected by vessels

Webcam: Measuring PPG signal from pupil and skin



Heart Rate Variability

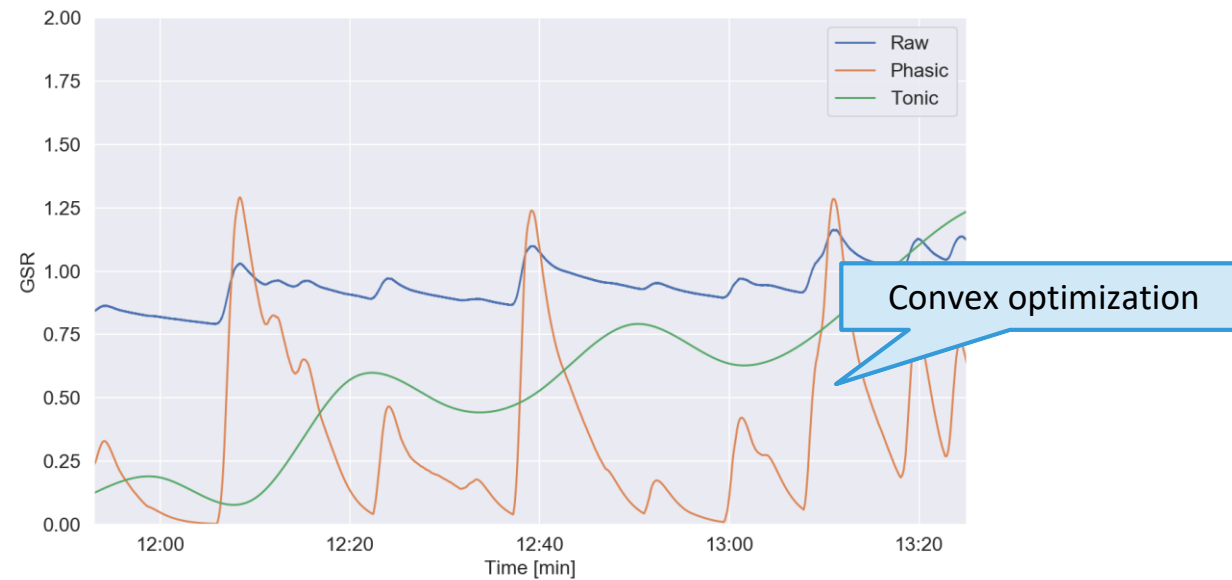
Time domain	LF: 0.04 – 0.1Hz	HF: 0.1 – 0.5Hz
	Frequency domain	
Standard deviation of R-R intervals	LF / (LF + HF)	
Difference between max and min R-R interval	LF / HF	
Percentage of successive R-R intervals that differ by more than 50 ms	LF and HF peak	
Root mean square of successive R-R interval differences		

Age, posture, physical condition & activity, breathing affect HRV

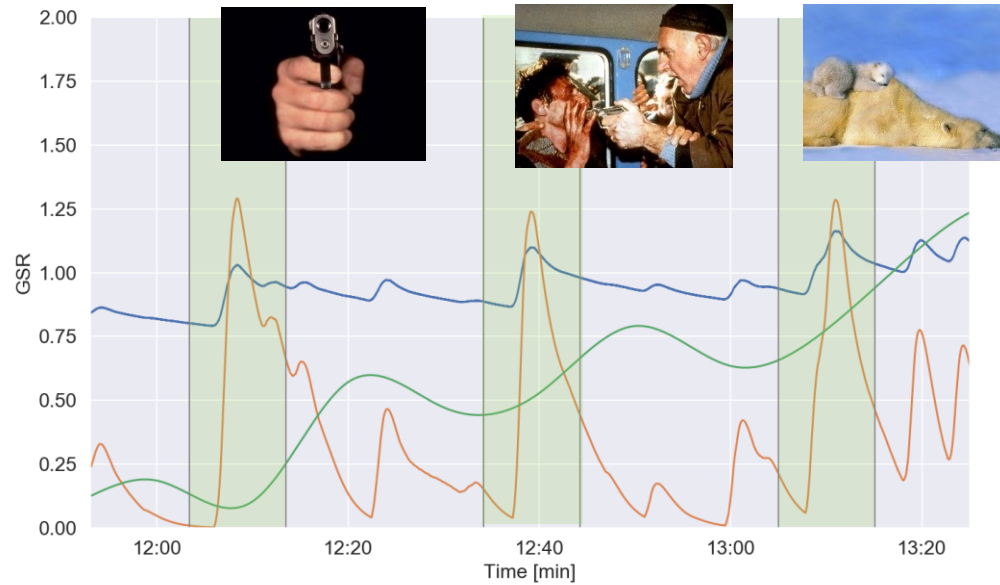
Indicative for arousal

Measures sweating level

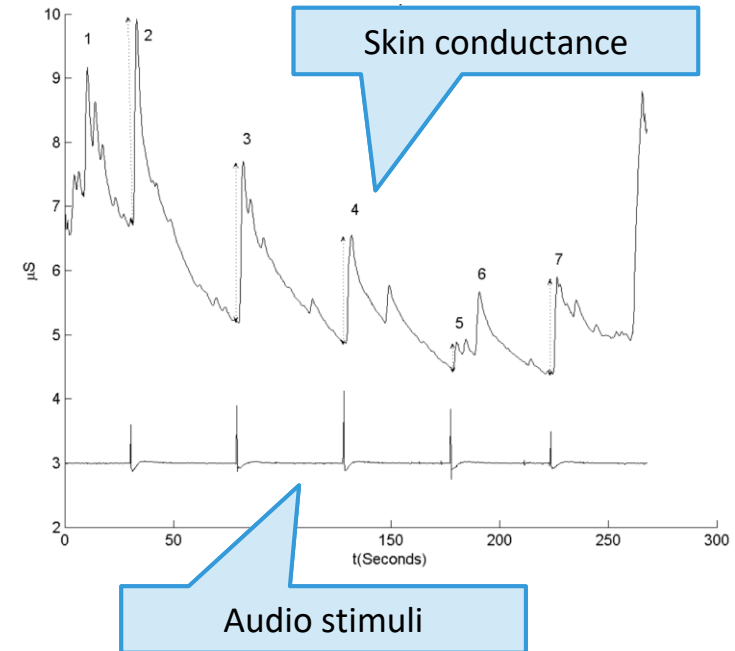
Used in lie detectors



Picture stimuli



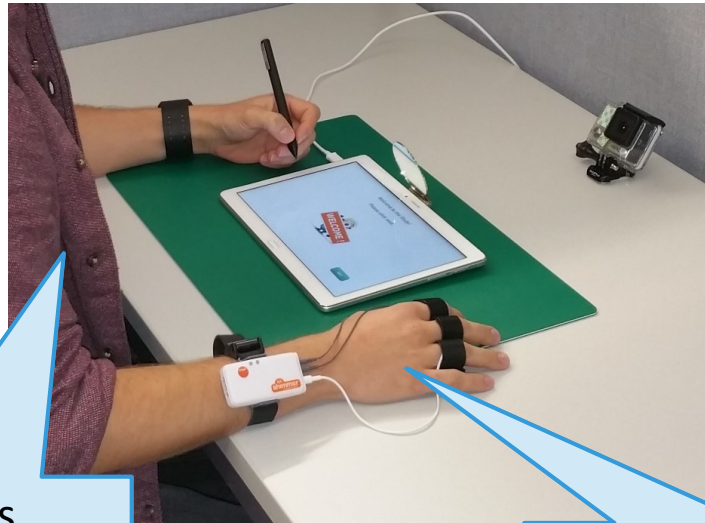
Audio stimuli





- 1 Amplitude
- 2 Latency
- 3 Rise time
- 4 Half-recovery time

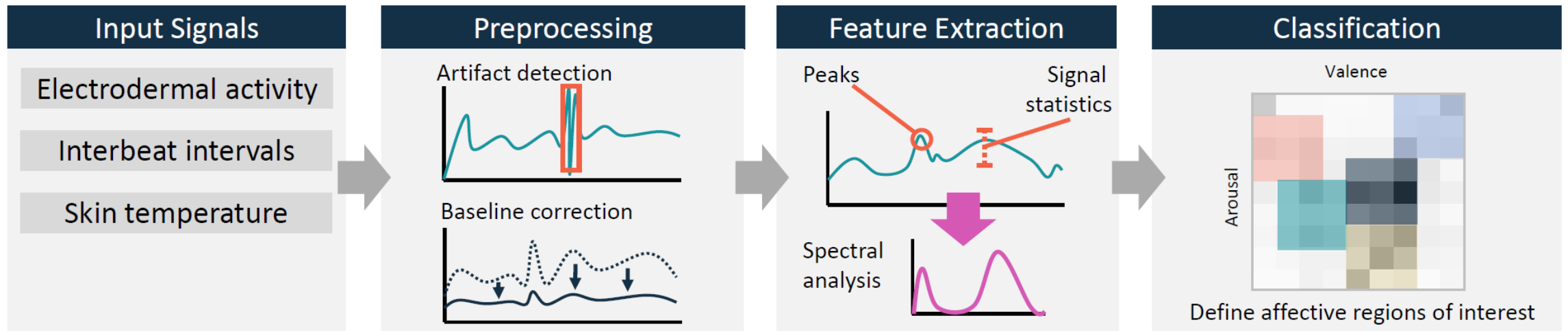
Motion affects skin conductance
Skin conductance response is not time invariant



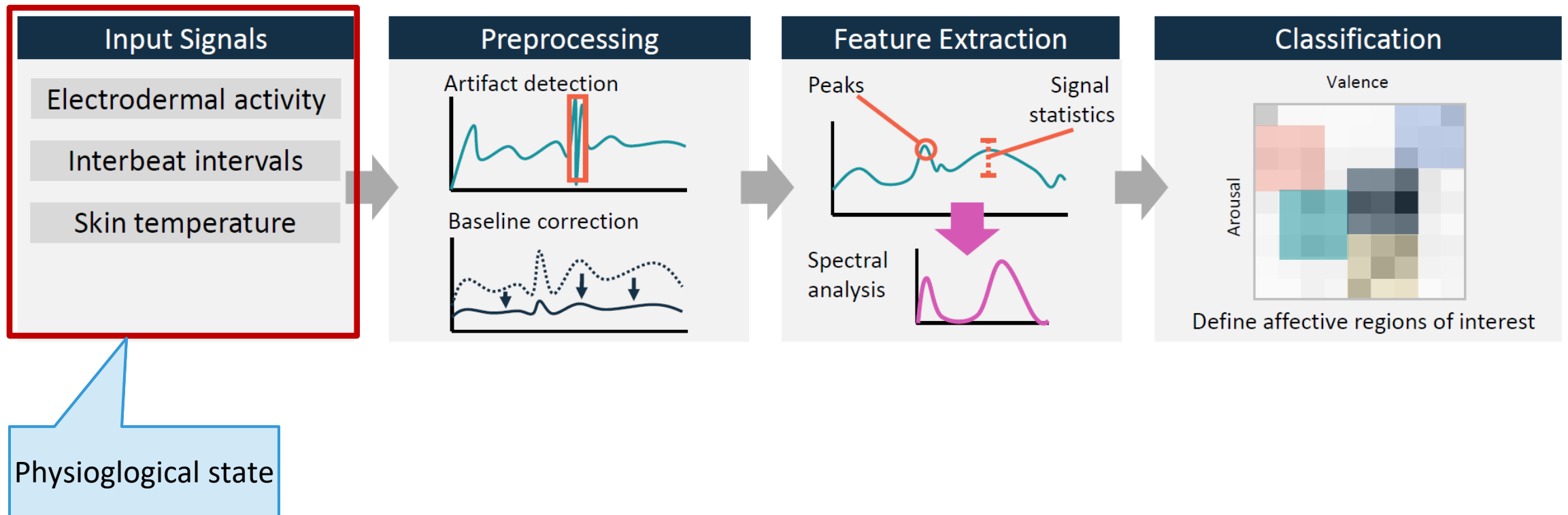
Heart measures
(Polar H10 chest strap)

Skin conductance
(Shimmer GSR+)

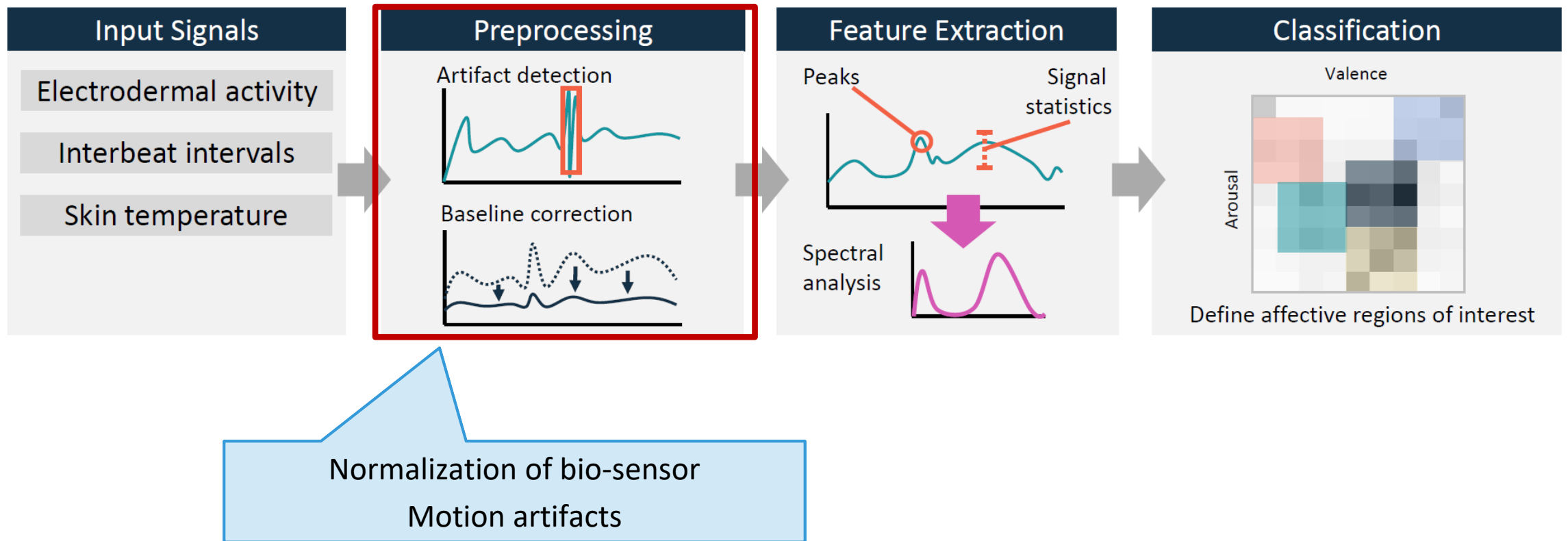
Biosensor Pipeline

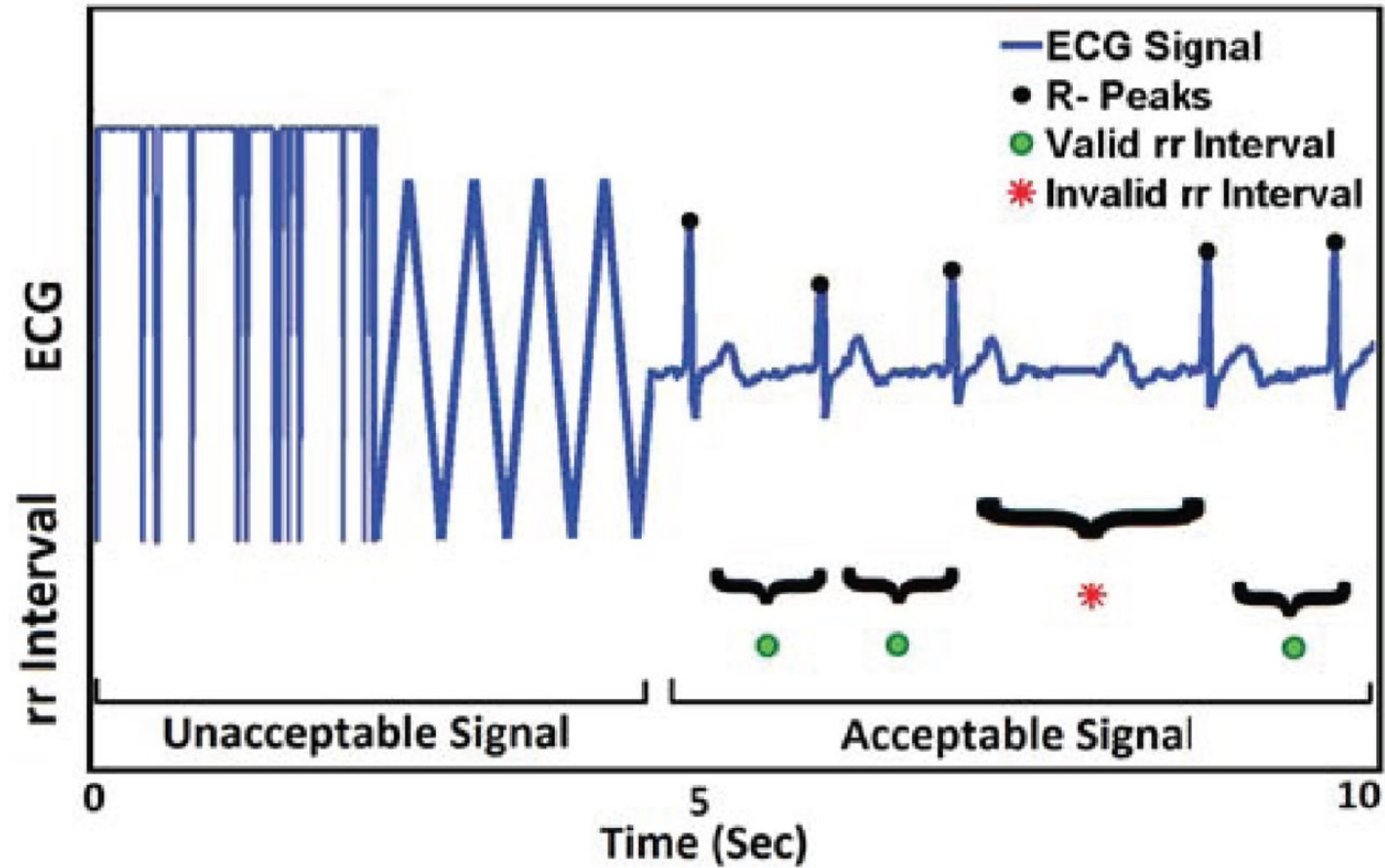


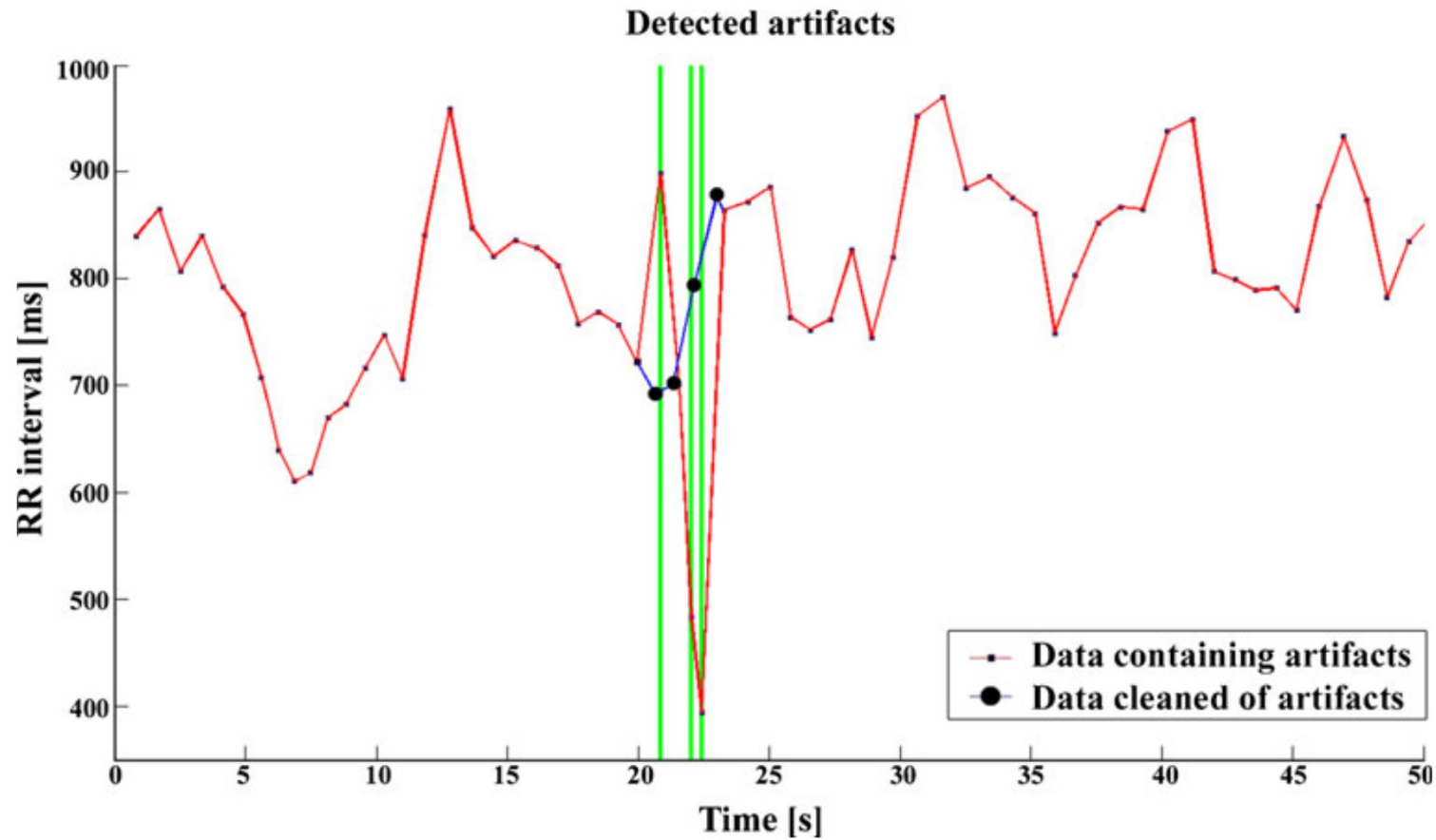
Biosensor Pipeline



Biosensor Pipeline







```
1 function isRRintValid ( $RR_n, RR_{n-1}, RR_{n+1}, RR_k$ );  
   Inputs :  $RR_n$ : current candidate R-R interval;  $RR_{n-1}$ :  
           previous candidate R-R interval;  $RR_{n+1}$ : next  
           candidate R-R interval;  $RR_k | k < n$ : last valid R-R  
           interval before  $RR_n$   
   Output:  $RR_n$  valid: is  $RR_n$  valid?  
2 if  $300ms \leq RR_{n-1} \leq 2000ms$  then  
3   if  $|RR_n - RR_{n-1}| \leq CBD$  then  
4      $RR_n$  valid = true;  
5   else  
6      $RR_n$  valid = false;  
7   end  
8 else  
9   if  $|RR_n - RR_k| \leq CBD$  then  
10     $RR_n$  valid = true;  
11  else  
12    if  $|RR_n - RR_{n-1}| \leq CBD$  and  
13       $|RR_n - RR_{n+1}| \leq CBD$  then  
14       $RR_n$  valid = true;  
15    else  
16       $RR_n$  valid = false;  
17    end  
18 end
```

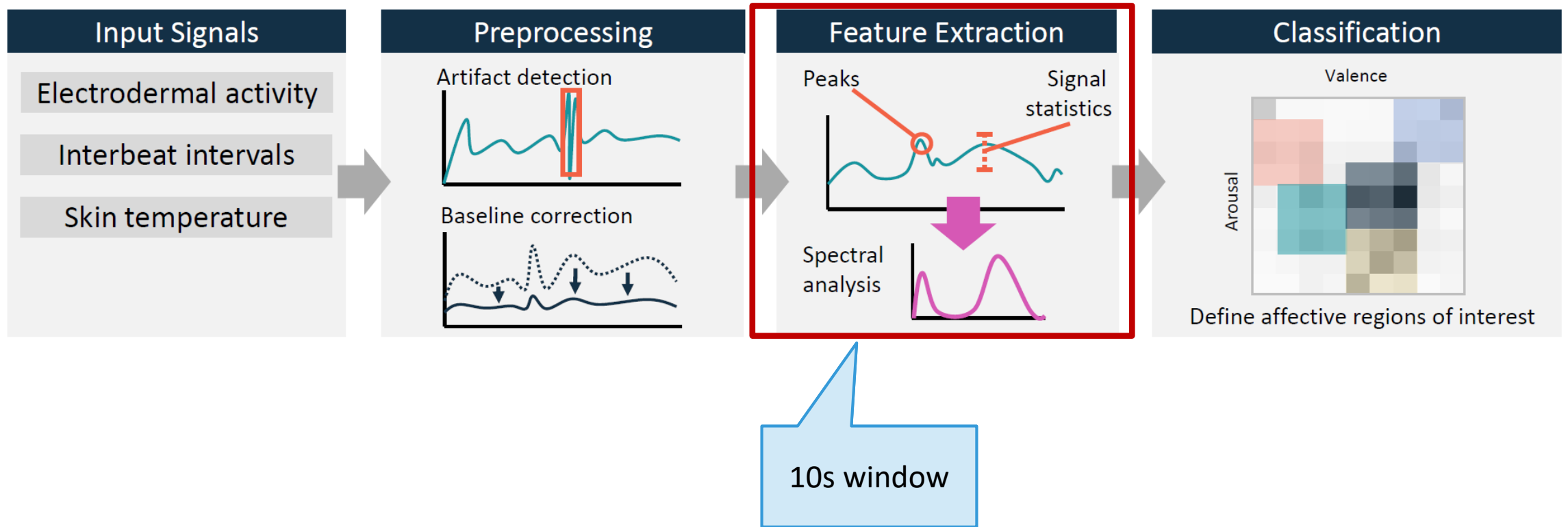
Hovsepian et al. 2015

Not relevant for exam

QD = Quartile Deviation = $(Q_3 - Q_1) / 2$
MAD = (Median Beat - 2.9 * QD) / 3
MED = 3.32 * QD

CBD = Criterion Beat Difference
CBD = (MAD + MED) / 2

Biosensor Pipeline

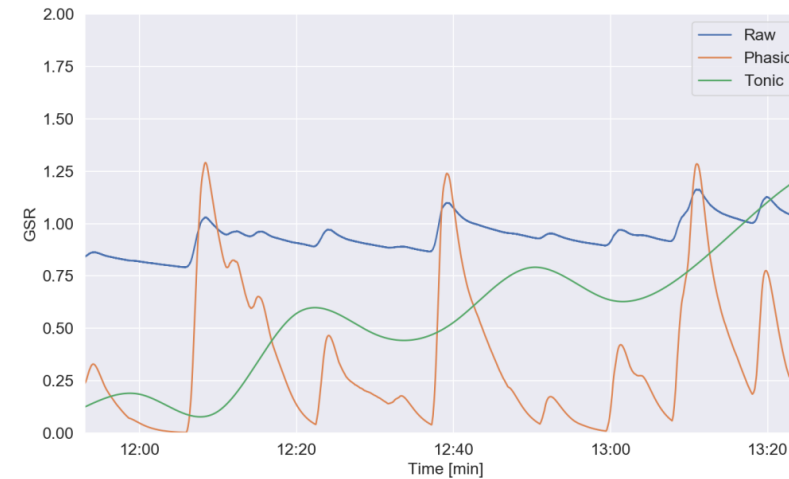


Heart Rate Variability

Time domain	LF: 0.04 – 0.1Hz	HF: 0.1 – 0.5Hz
Frequency domain	LF / (LF + HF)	
Standard deviation of R-R intervals	LF / HF	
Difference between max and min R-R interval	LF and HF peak	
Percentage of successive R-R intervals that differ by more than 50 ms		
Root mean square of successive R-R interval differences		

Python: cvxEDA library
(<https://github.com/lciti/cvxEDA>)

Artifact detection not necessary
(Implicit filtering)



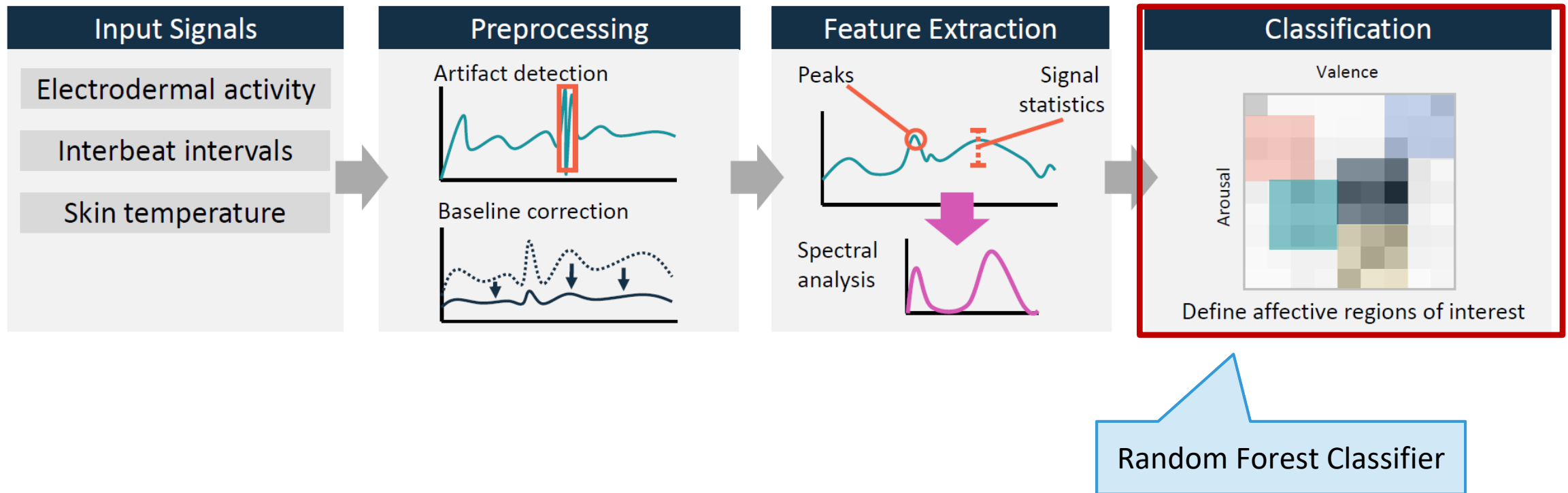
Phasic & Tonic

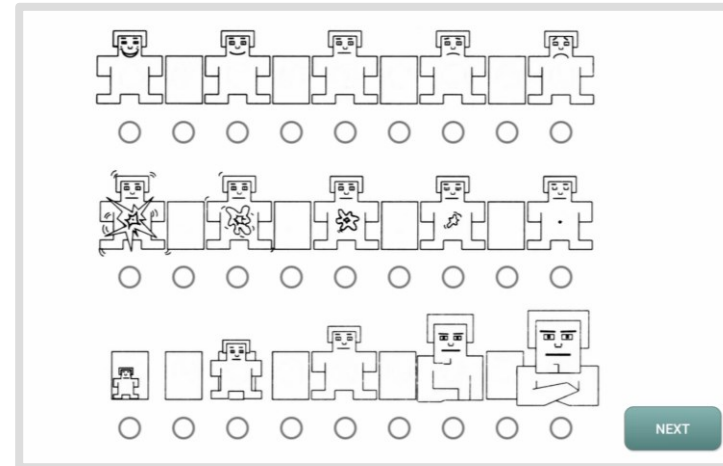
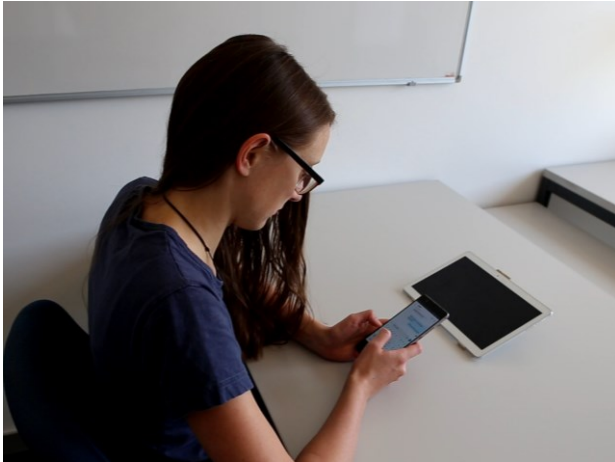
Mean
Standard deviation
Min
Max
Slope

Phasic

Area under the curve
Peaks
Amplitude statistics (mean, min, max)

Biosensor Pipeline





Valence

Arousal

Dominance

Basic Emotions & Stress

The Self-Assessment Manikin (SAM) interface consists of three rows of manikin faces, each with a corresponding radio button below it. The first row, labeled 'Valence', has five faces ranging from neutral to happy, with the third face selected. The second row, labeled 'Arousal', has five faces ranging from calm to excited, with the third face selected. The third row, labeled 'Dominance', has five faces ranging from passive to assertive, with the third face selected. A 'NEXT' button is located at the bottom right of the interface.

Self-Assessment Manikin

Valence

Arousal

Dominance

The Basic Emotions & Stress interface displays four basic emotions: Angry, Sad, Happy, and Surprise, each with a corresponding emoji and label. Below these is a 'Stressed' icon consisting of three blue droplets. A 'NEXT' button is located at the bottom right of the interface.

Basic Emotions & Stress

Electroencephalogram



Electromyogram



Respiration



Emotion Theories

- James-Lange
- Cannon-Bard
- Schachter Singer
- Dimensional
- Basic Emotions

Biosensors

- Heart Rate
- Skin Conductance
- Random Forest

Video

- Action Units
- Eye Blinks / Gaze
- Mouth Aspect Ratio
- Fidgeting

- **Oxford Handbook of Affective Computing – Chapter 3, 5, 8, 10, 14**
- Wampfler, Rafael, et al. "Affective state prediction in a mobile setting using wearable biometric sensors and stylus." *Proceedings of The 12th International Conference on Educational Data Mining (EDM 2019)*. 2019.
- Wampfler, Rafael, et al. "Affective State Prediction Based on Semi-Supervised Learning from Smartphone Touch Data." *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 2020.
- Wampfler, Rafael, et al. "Image Reconstruction of Tablet Front Camera Recordings in Educational Settings." *International Educational Data Mining Society* (2020).

- Loewenstein, George, and Jennifer S. Lerner: The role of affect in decision making.
- Whiten, Andrew (Ed.) Natural Theories of Mind: Chapter 1, Chapter 2, Chapter 3.
- Rosalind Picard: *Affective Computing* Introduction, Chapters 1, 2 and 5-7
- Antonio Damasio: *Descartes' Error* Chapters 1 and 3
- Daniel Kahneman: *Thinking, Fast and Slow* Chapters 1-3
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